南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

计算机组成原理
COMPUTER ORGANIZATION

zhangj4@sustech.edu.cn

Computer Organization

Lecture 6

Arithmetic Operations on Integers

zhangj4@sustech.edu.cn

Recap

- Computer Performance
  - Response time vs. throughput
  - Performance = 1 / Execution Time
  - Notions: clock frequency (f) / clock period (Tc) / clock cycle
- Execution Time
  - Elapsed time
  - CPU time = IC ✕ CPI ✕ Tc = IC ✕ CPI / f
  - Notions: instruction count (IC) / cycles per instruction (CPI)
- Power consumption
  - Power = ½ ✕ C ✕ V^2 ✕ f
- Performance comparison
  - SPEC CPU benchmark, geometric mean
- Amdahl's law: $T_{improved}=T_{affected}/factor + T_{unaffected}$

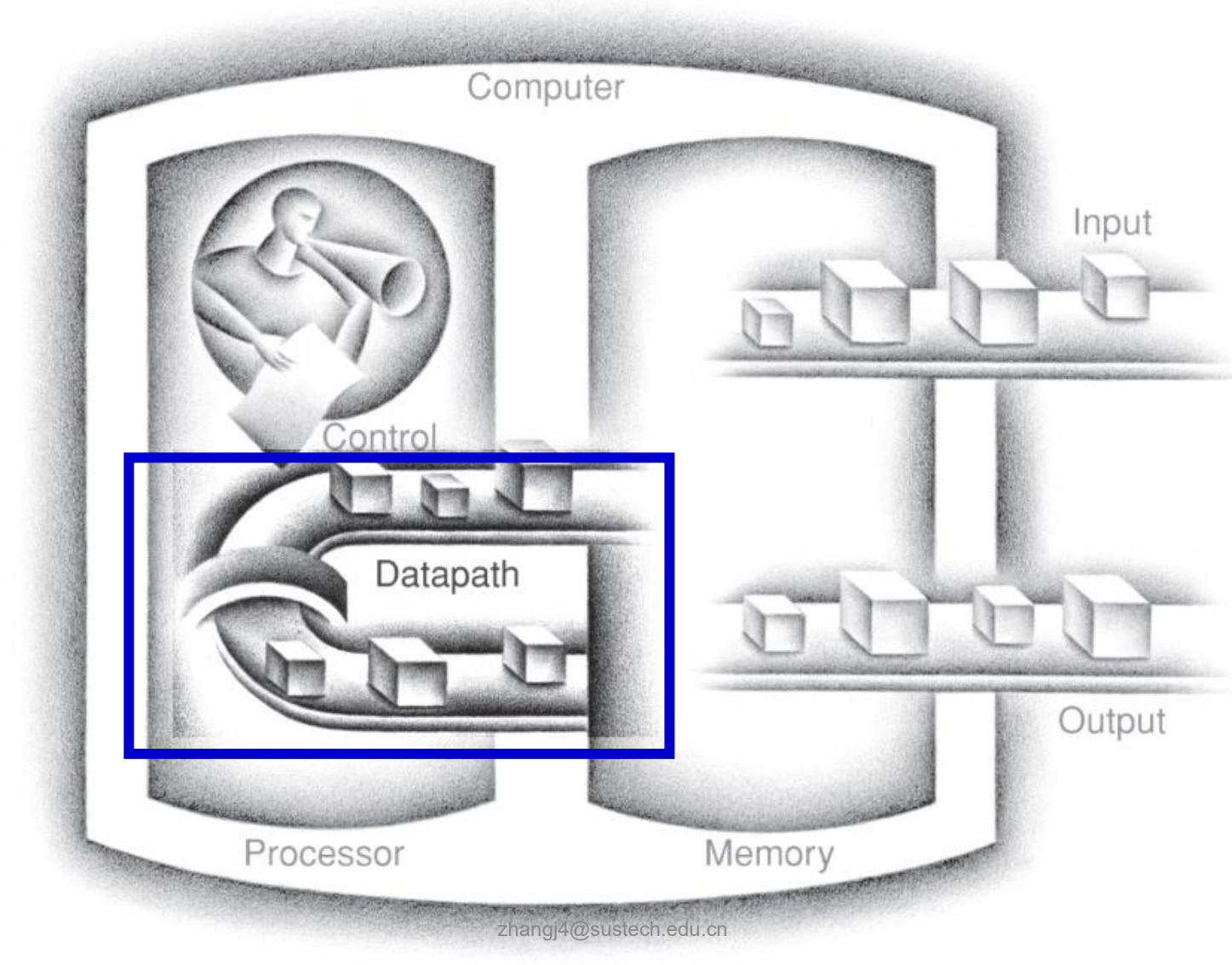
zhangj4@sustech.edu.cn

Outline

- Integer addition
- Integer subtraction
- Integer multiplication
- Integer division

zhangj4@sustech.edu.cn

Datapath



Integer Addition

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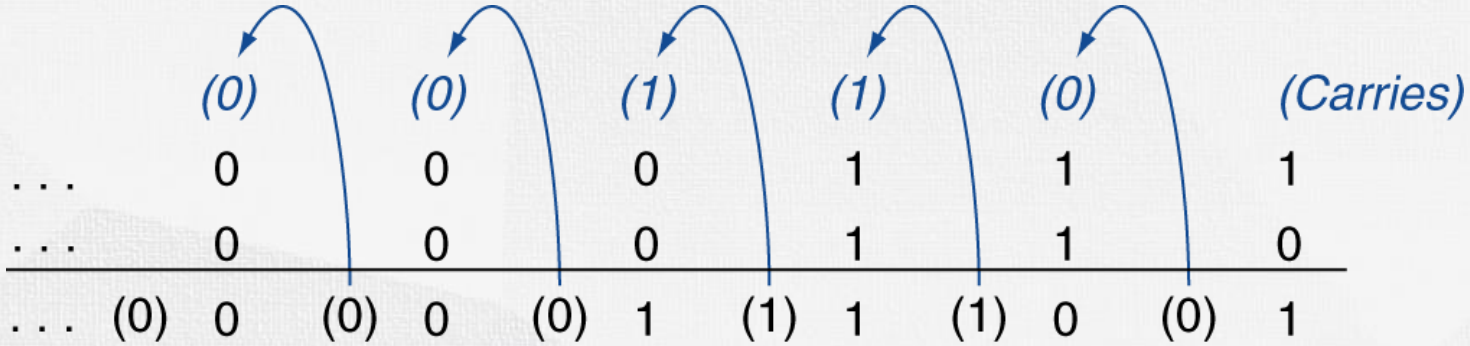
Integer Addition

- Example: 7 + 6

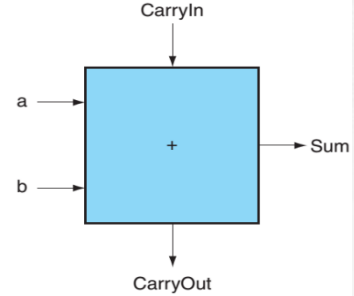


- Overflow
  - please write down the 8-bit signed integer addition:
  - $0100\ 0000_{two}+0100\ 0000_{two}=?$
  - $1000\ 0000_{two}+1000\ 0000_{two}=?$
  - $0100\ 0000_{two}+1100\ 0000_{two}=?$
  - $1100\ 0000_{two}+1100\ 0000_{two}=?$
  - Please write down the equations in binary and decimal.

Overflow

- Examples in last page:
  - 8-bit signed integer range: -128 ~ 127
  - $0100\ 0000_{two}+0100\ 0000_{two}=1000\ 0000_{two}$ 64 + 64= -128 Overflow
  - $1000\ 0000_{two}+1000\ 0000_{two}=(1)0000\ 0000_{two}$ -128 + (-128) = 0 Overflow
  - $0100\ 0000_{two}+1100\ 0000_{two}=(1)0000\ 0000_{two}$ 64+ (-64)= 0 No overflow
  - $1100\ 0000_{two}+1100\ 0000_{two}=(1)1000\ 0000_{two}$ -64+(-64)=-128 No overflow
- Overflow if result out of range
  - no overflow, if ddding +ve(positive) and –ve(negative) operands
  - Overflow, if:
    - Adding two +ve operands, get –ve operand
    - Adding two -ve operands, get +ve operand

1-bit adder



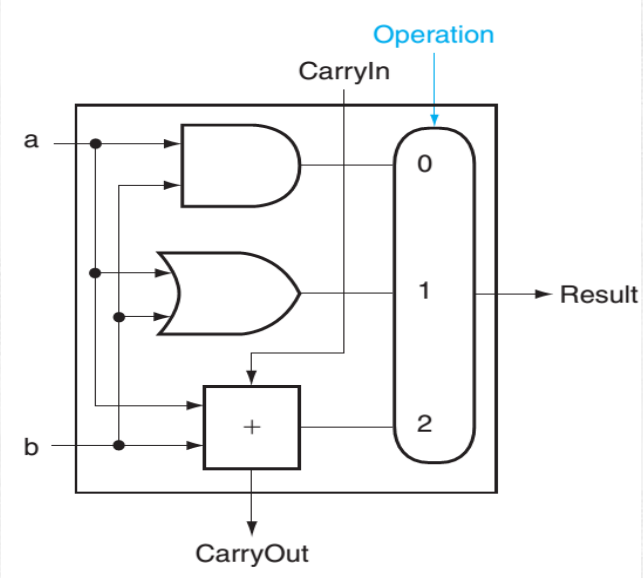
$$\text{Sum} = (a \cdot \overline{b} \cdot \overline{\text{CarryIn}}) + (\overline{a} \cdot b \cdot \overline{\text{CarryIn}}) + (\overline{a} \cdot \overline{b} \cdot \text{CarryIn}) + (a \cdot b \cdot \text{CarryIn})$$

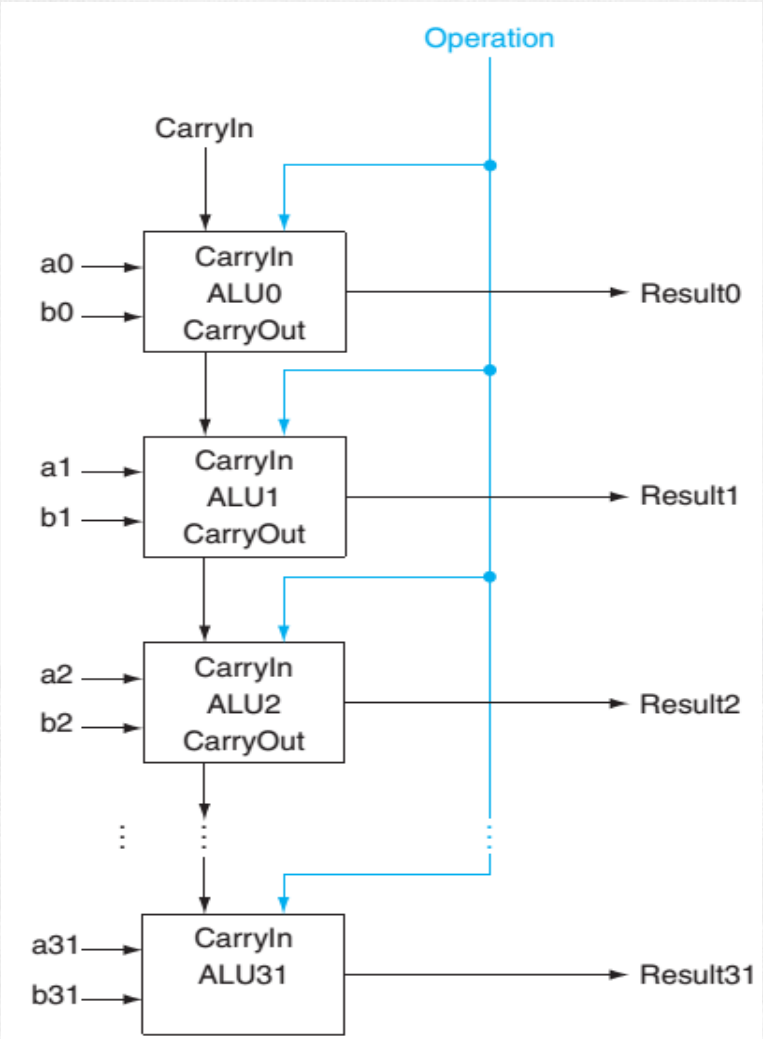
$$\text{CarryOut} = (b \cdot \text{CarryIn}) + (a \cdot \text{CarryIn}) + (a \cdot b)$$

Inputs			Outputs		
a	b	CarryIn	CarryOut	Sum	Comments
0	0	0	0	0	$0 + 0 + 0 = 00_{two}$
0	0	1	0	1	$0 + 0 + 1 = 01_{two}$
0	1	0	0	1	$0 + 1 + 0 = 01_{two}$
0	1	1	1	0	$0 + 1 + 1 = 10_{two}$
1	0	0	0	1	$1 + 0 + 0 = 01_{two}$
1	0	1	1	0	$1 + 0 + 1 = 10_{two}$
1	1	0	1	0	$1 + 1 + 0 = 10_{two}$
1	1	1	1	1	$1 + 1 + 1 = 11_{two}$

1-bit ALU and 32-bit ALU

- ALU: arithmetic logical unit
- 1-bit ALU and 32-bit ALU
  - If op = 0, o = a & b (and)
  - If op = 1, o = a | b (or)
  - If op = 2, o = a + b (add)





zhangj4@sustech.edu.cn

Integer Subtraction

zhangj4@sustech.edu.cn

Integer Subtraction

- Add negation of second operand
- Example: 7 – 6 = 7 + (–6)

  +7: 0000 0000 … 0000 0111

  –6: 1111 1111 … 1111 1010

  +1: 0000 0000 … 0000 0001

- Overflow if result out of range
  - No overflow, if subtracting two +ve or two –ve operands
  - Overflow, if:
    - Subtracting +ve from –ve operand, and the result sign is 0 (+ve)
    - Subtracting –ve from +ve operand, and the result sign is 1 (-ve)

zhangj4@sustech.edu.cn

Overflow Situations

Operation	Operand A	Operand B	Result indicating overflow
$A + B$	≥ 0	≥ 0	< 0
$A + B$	< 0	< 0	≥ 0
$A - B$	≥ 0	< 0	< 0
$A - B$	< 0	≥ 0	≥ 0

- Example:
  - 12+3=15

```
  00001100
+ 00000011
----------
  00001111
```

120+15=135

```
  01111000
+ 00001111
----------
  10000111
```

12-3=9

12-3 = 12+(-3)

```
  00001100
+ 11111101
----------
  00001001
```

-100-50=-150

-100-50=-100+(-50)

```
  10011100
+ 11001110
----------
  01101010
```

Overflow

Arithmetic for Multimedia – Saturating Operation

- Graphics and media processing operates on vectors of 8-bit and 16-bit data
  - Use 64-bit adder, with partitioned carry chain
    - Operate on 8×8-bit, 4×16-bit, or 2×32-bit vectors
  - SIMD (single-instruction, multiple-data)
- Saturating operations
  - On overflow, result is largest representable value
    - c.f. 2s-complement modulo arithmetic
    - E.g. a=200, b=200, both are 8-bit unsigned integers (range: 0-255)
    - For regular addition: a+b=400 mod 256 = 144, for saturating addition: a+b=255
  - E.g., change the volume and brightness in audio or video

Integer Multiplication

zhangj4@sustech.edu.cn

Multiplication

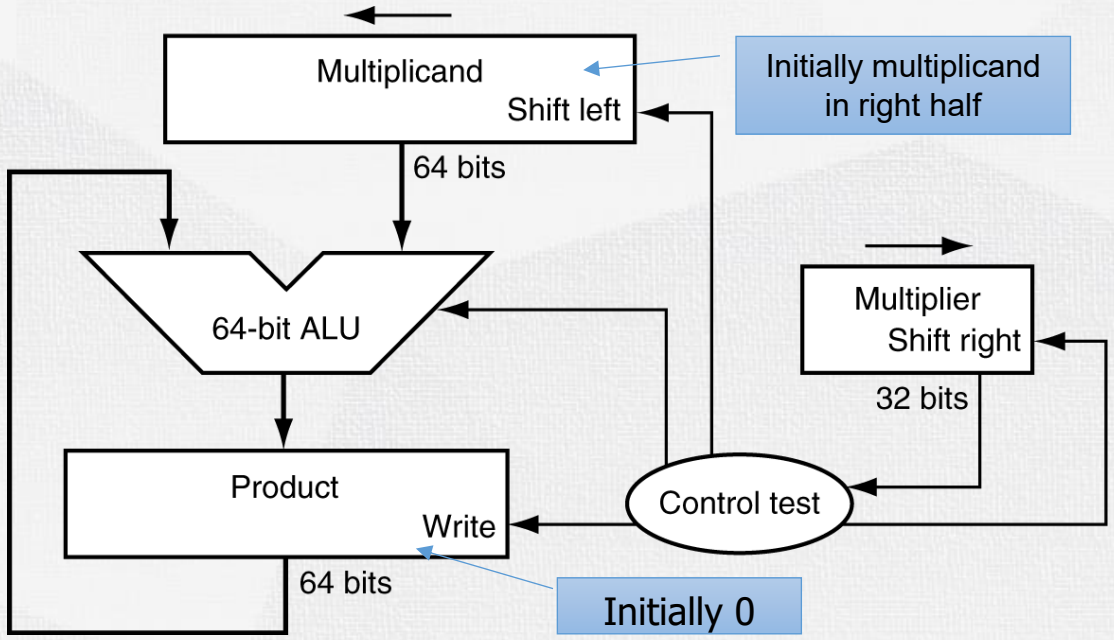
- In every step
  - multiplicand is shifted
  - next bit of multiplier is examined (also a shifting step)
  - if this bit is 1, shifted multiplicand is added to the product

multiplicand

multiplier

product

```
   1000
×  1001
   1000
  0000
 0000
1000
1001000
```

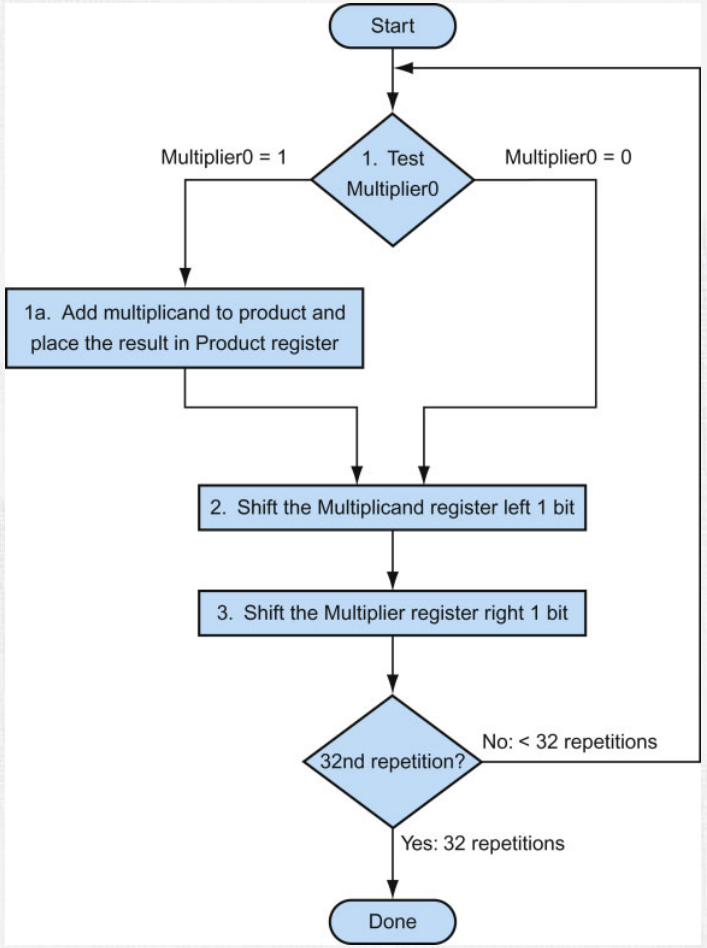


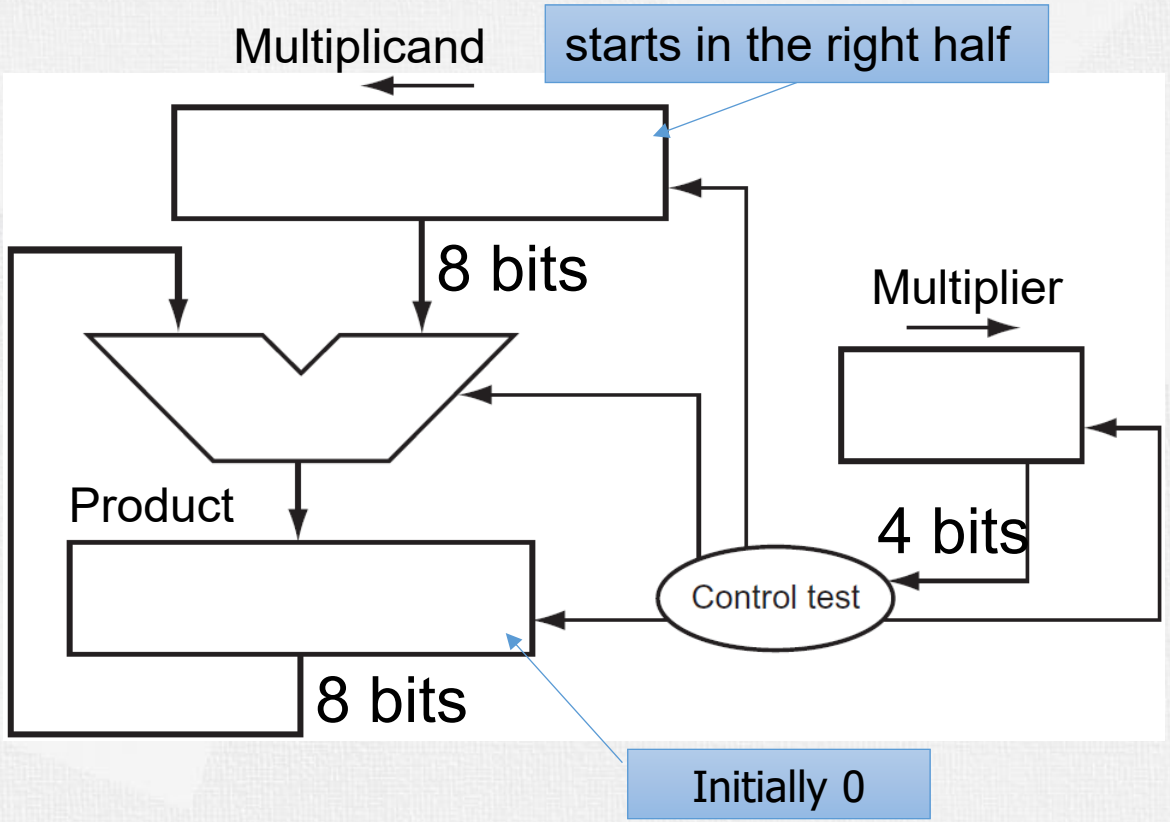
Length of product is the sum of operand lengths

zhangj4@sustech.edu.cn

Multiplication Example

- Multiply 8_{ten} (1000_{two}) by 9_{ten} (1001_{two})
  - How values change in Mcand, Mplier and Product Registers?

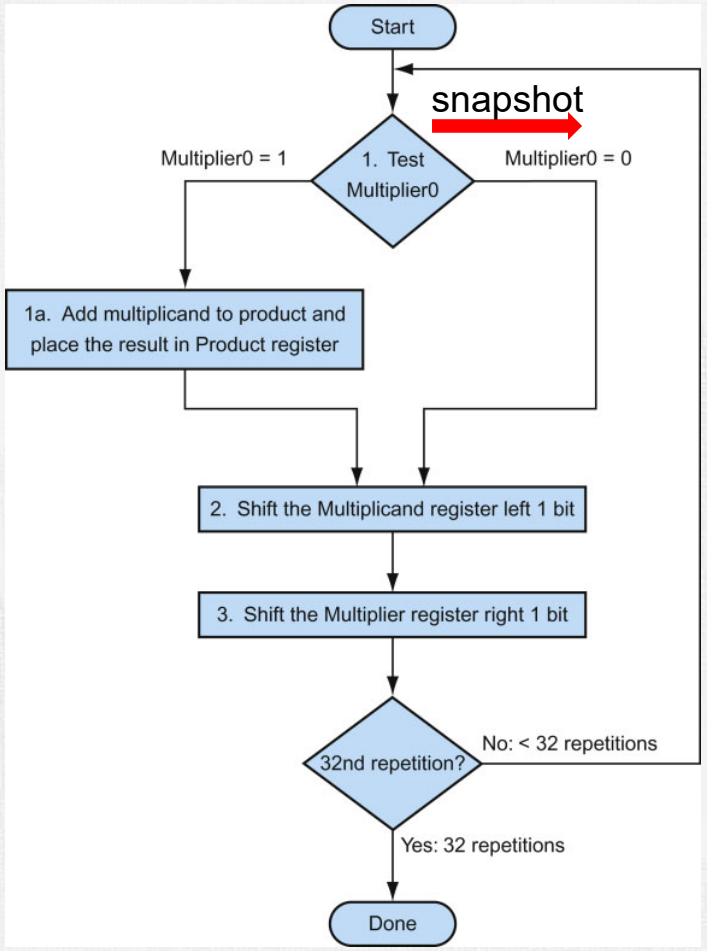


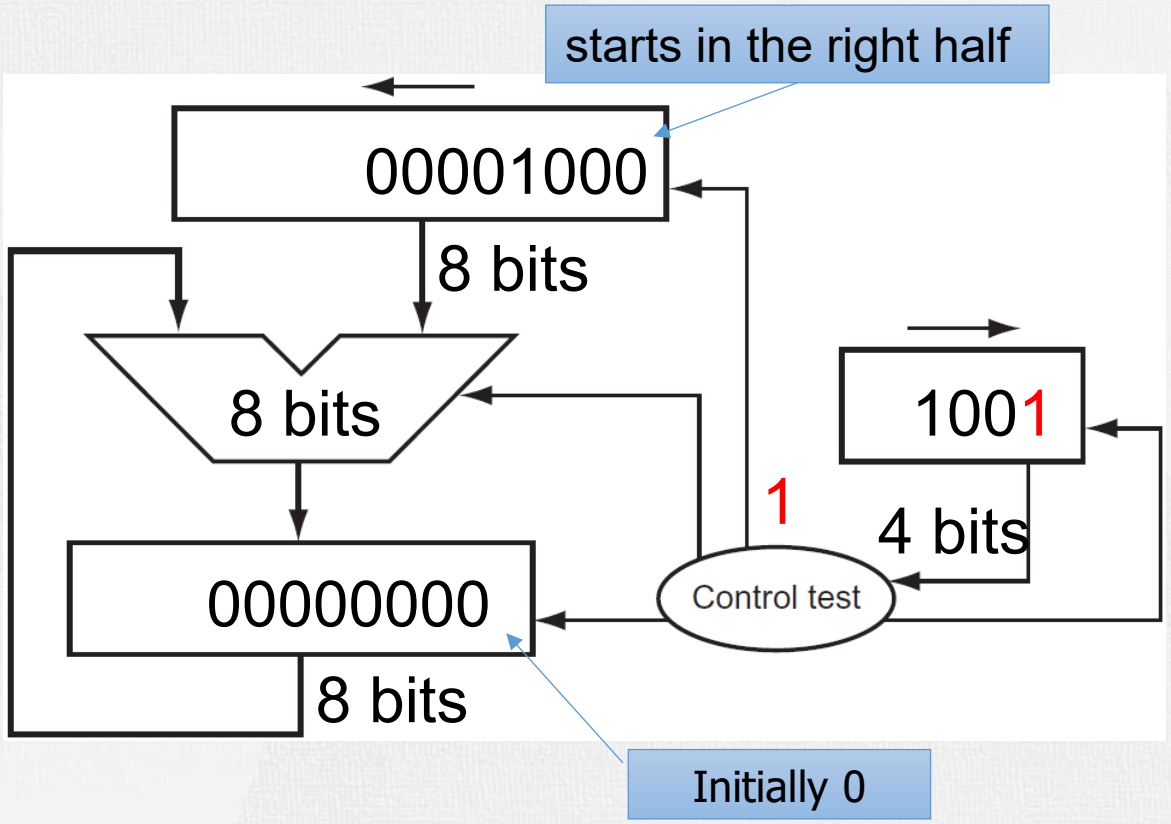


zhangj4@sustech.edu.cn

Multiplication Example

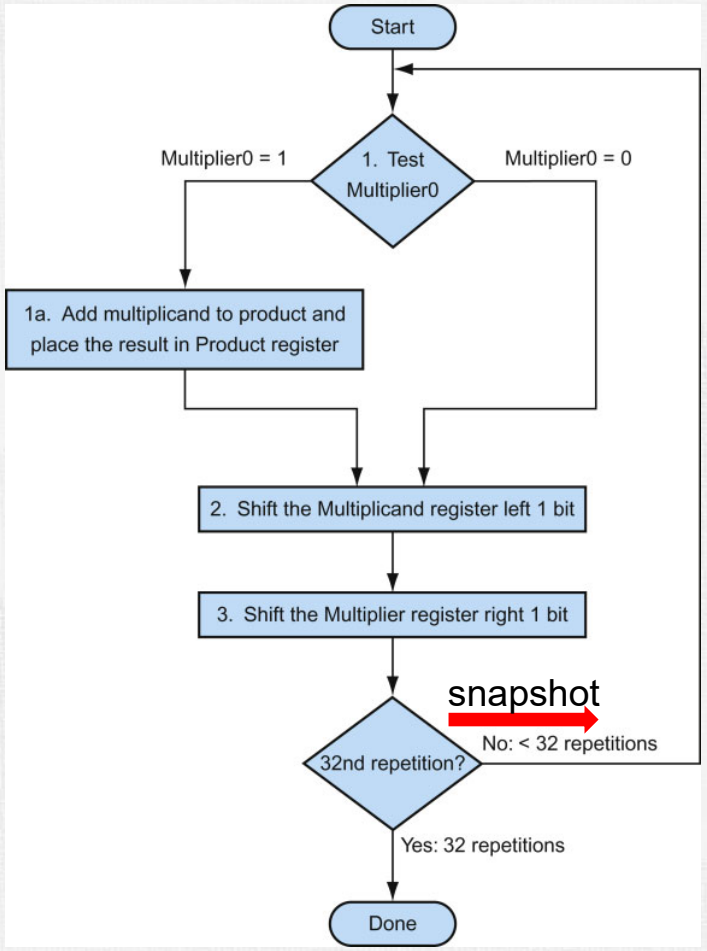
- Multiply 8_{ten} (1000_{two}) by 9_{ten} (1001_{two})

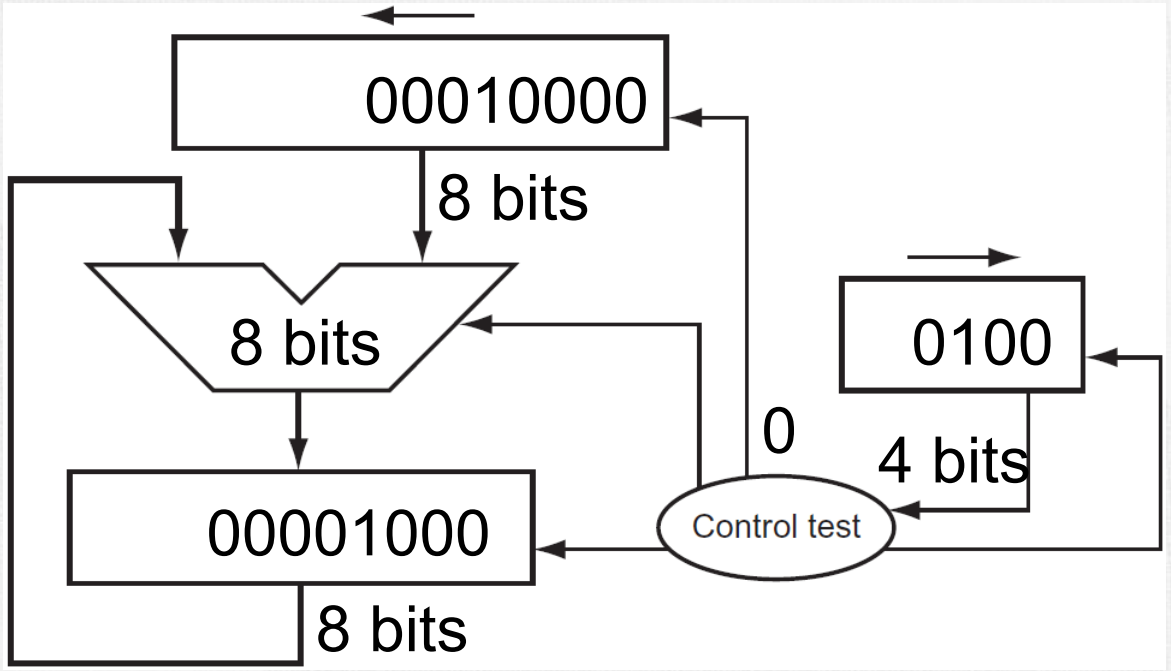




Multiplication Example

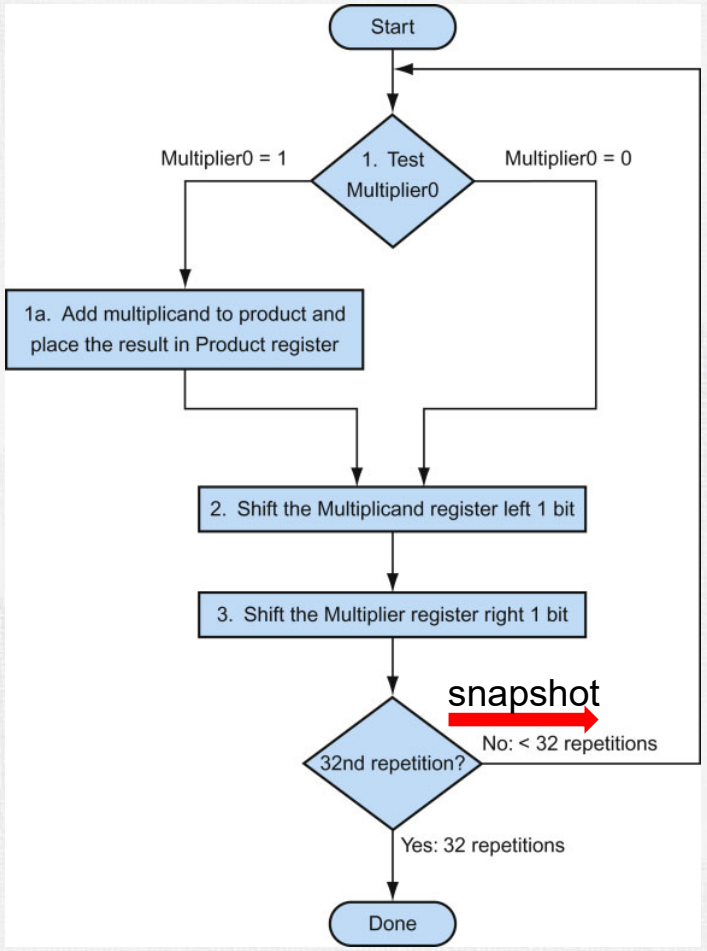
- Multiply 8_{ten} (1000_{two}) by 9_{ten} (1001_{two})

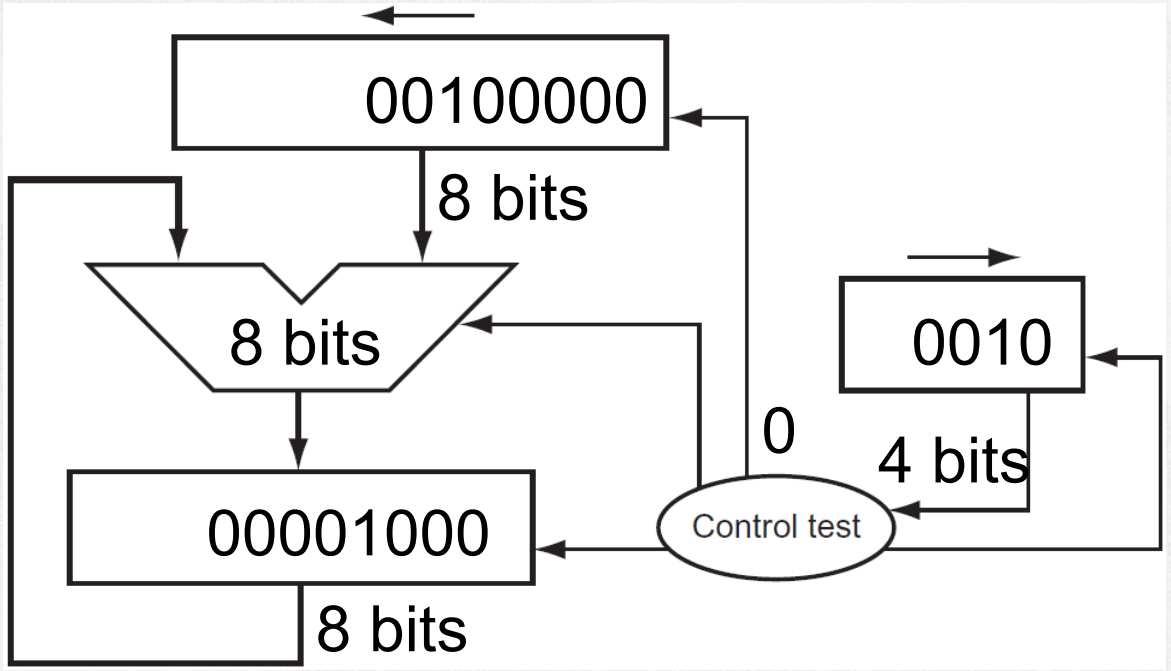




Multiplication Example

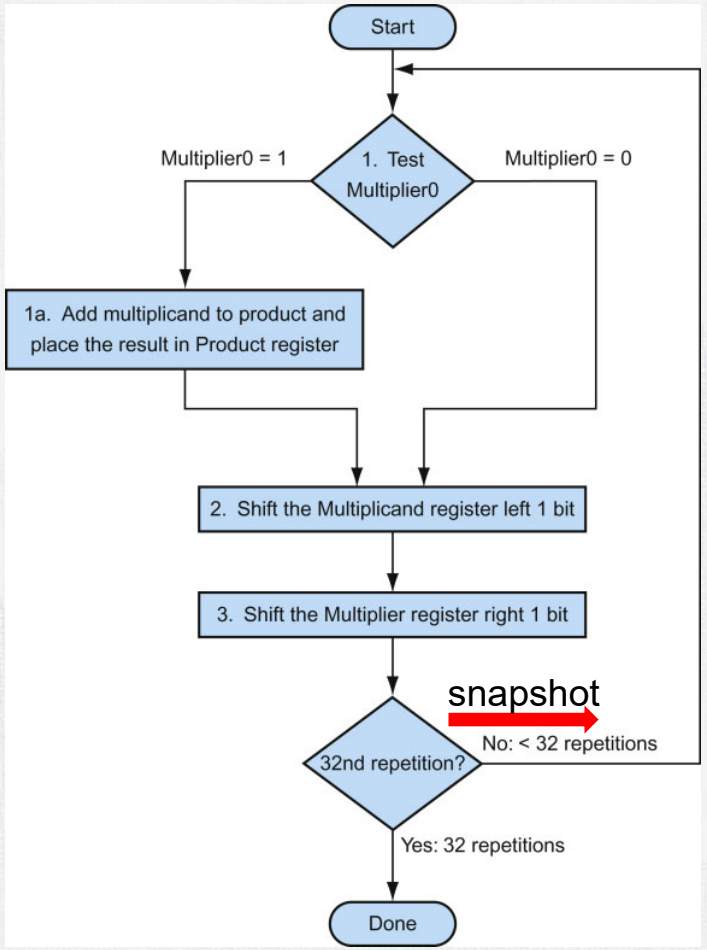
- Multiply 8_{ten} (1000_{two}) by 9_{ten} (1001_{two})

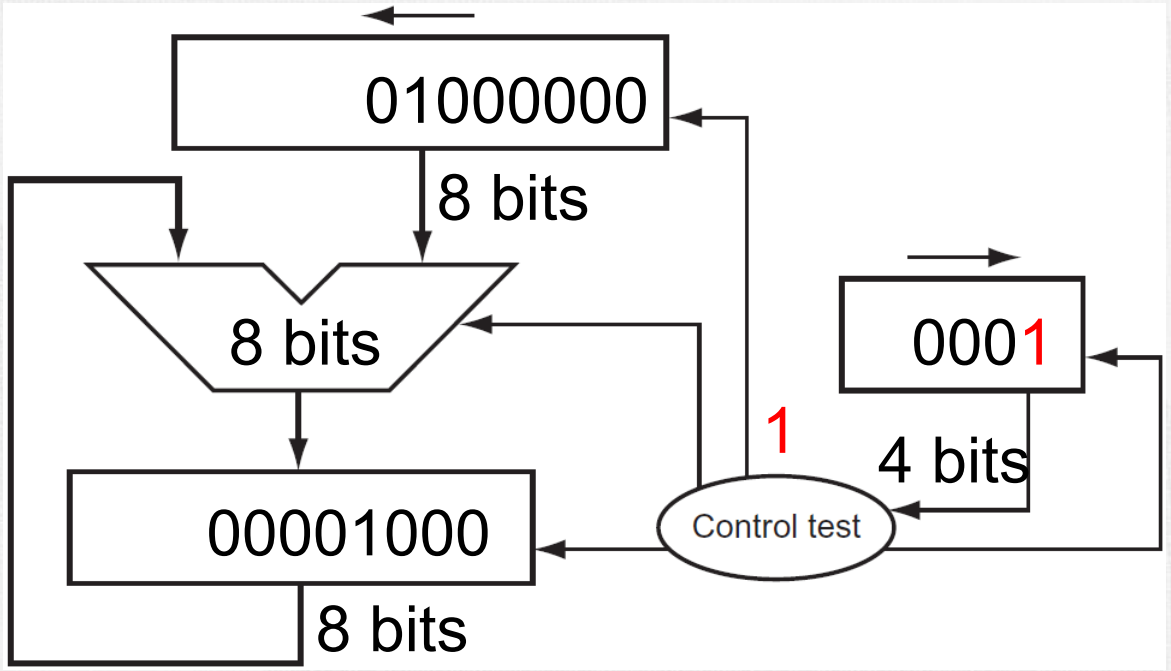




Multiplication Example

- Multiply 8_{ten} (1000_{two}) by 9_{ten} (1001_{two})





Multiplication Example

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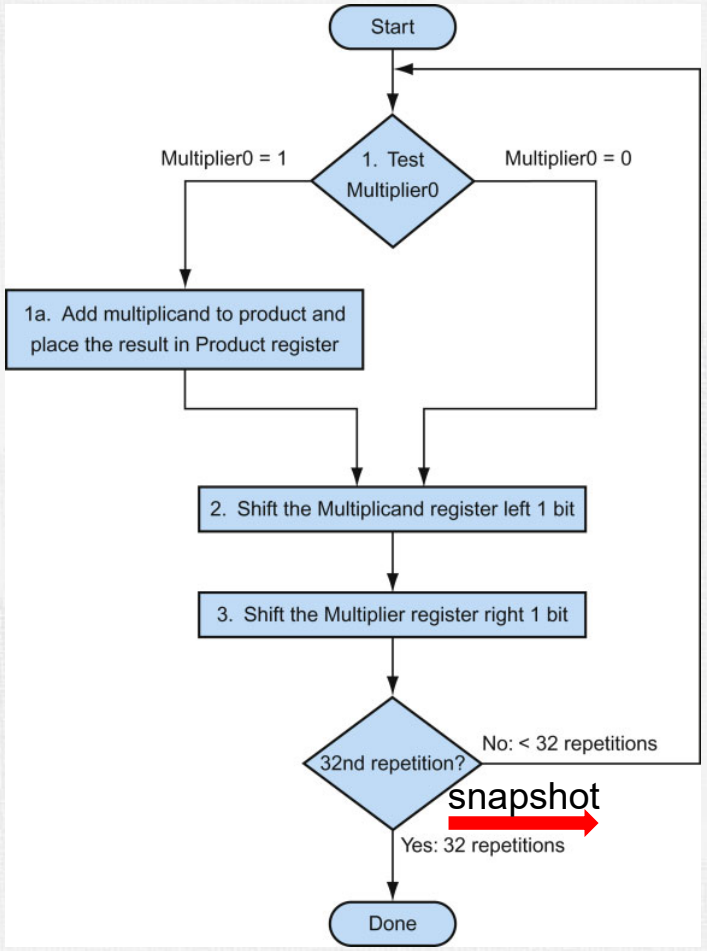
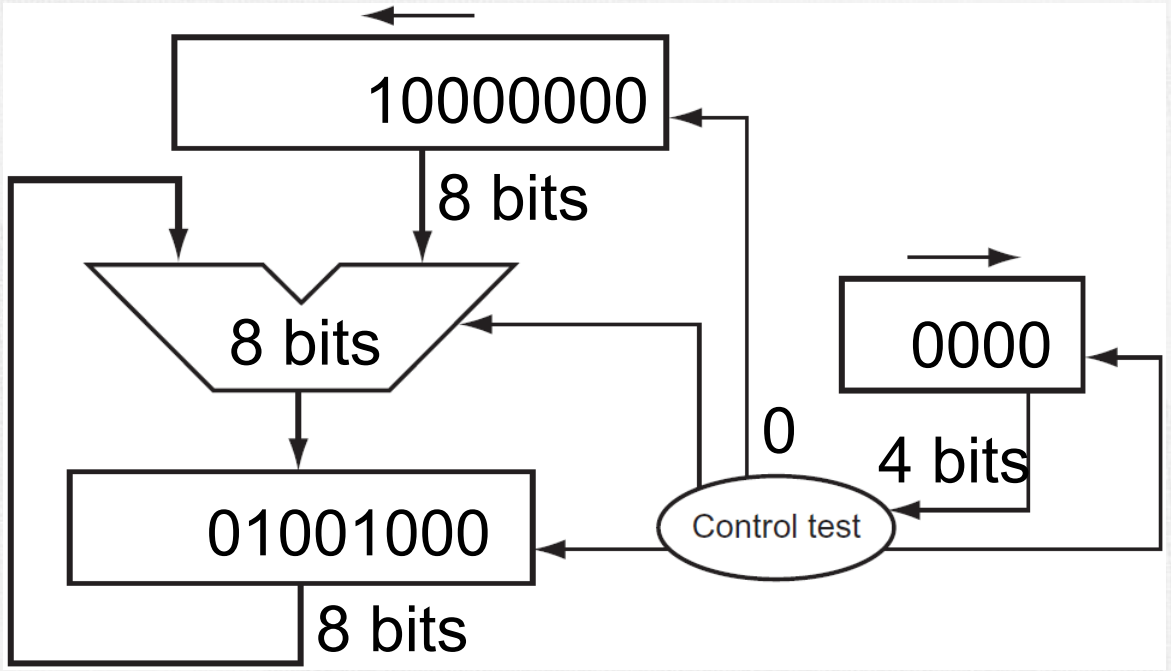




Done!

zhangj4@sustech.edu.cn

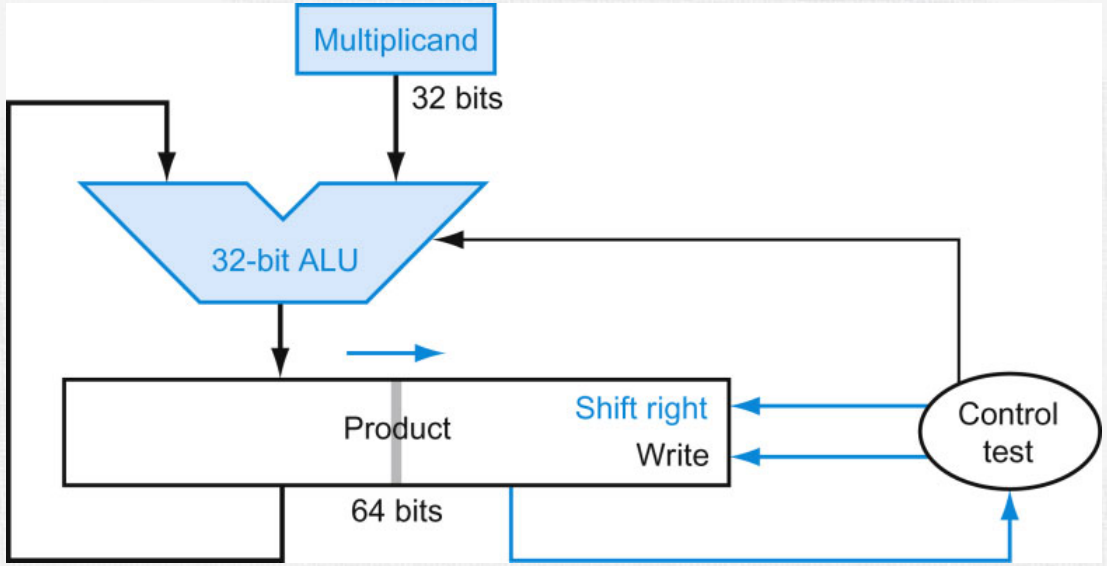
Multiplication Example

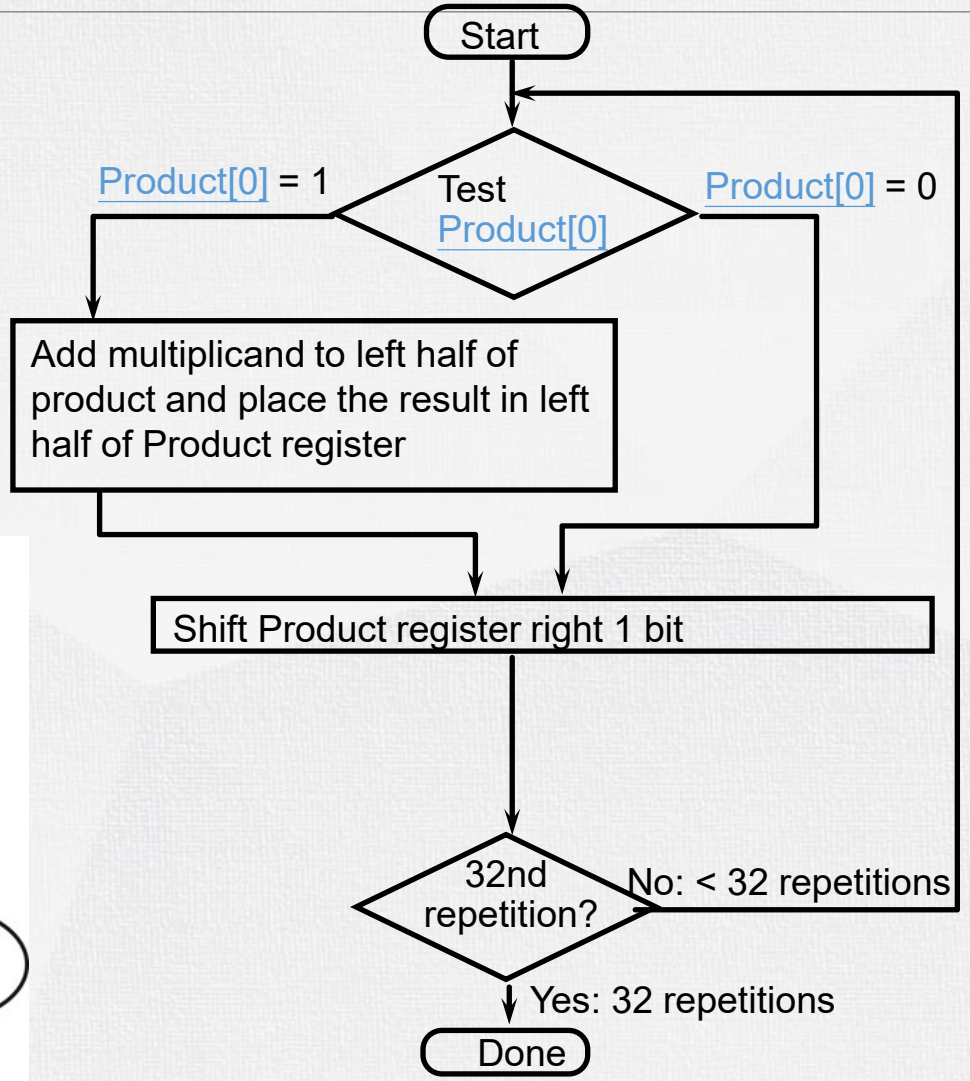
- Multiply 8_{ten} (1000_{two}) by 9_{ten} (1001_{two})

Iter	Step	Multiplier	Multiplicand	Product
0	Initial values	1001	0000 1000	0000 0000
1	1 ⇒ Prod = Prod + Mcand	1001	0000 1000	0000 1000
	Shift left Multiplicand	1001	0001 0000	0000 1000
	Shift right Multiplier	0100	0001 0000	0000 1000
2	0 ⇒ No operation	0100	0001 0000	0000 1000
	Shift left Multiplicand	0100	0010 0000	0000 1000
	Shift right Multiplier	0010	0010 0000	0000 1000
3	Same steps as 2	0010	0010 0000	0000 1000
		0010	0100 0000	0000 1000
		0001	0100 0000	0000 1000
4	Same steps as 1	0001	0100 0000	0100 1000
		0001	1000 0000	0100 1000
		0000	1000 0000	0100 1000

Optimized Multiplier Hardware

- Multiplier initially in right half of product register, 32-bit ALU and multiplicand is untouched
- Check the 0th bit in Product register, if 1, add left half of product with multiplicand
- The sum keeps shifting right, at every step, number of bits in product + multiplier = 64

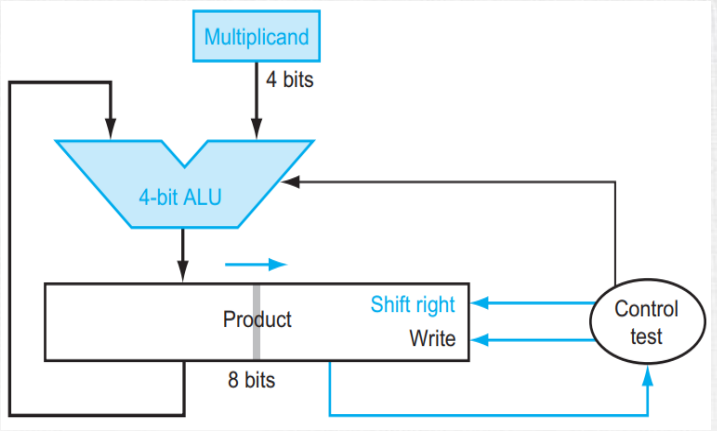




zhangj4@sustech.edu.cn

Optimized Multiplier

- Example:
  - Multiply 2_{ten} (0010_{two}) by 7_{ten} (0111_{two})
  - result = 00001110_{two} (14_{ten})

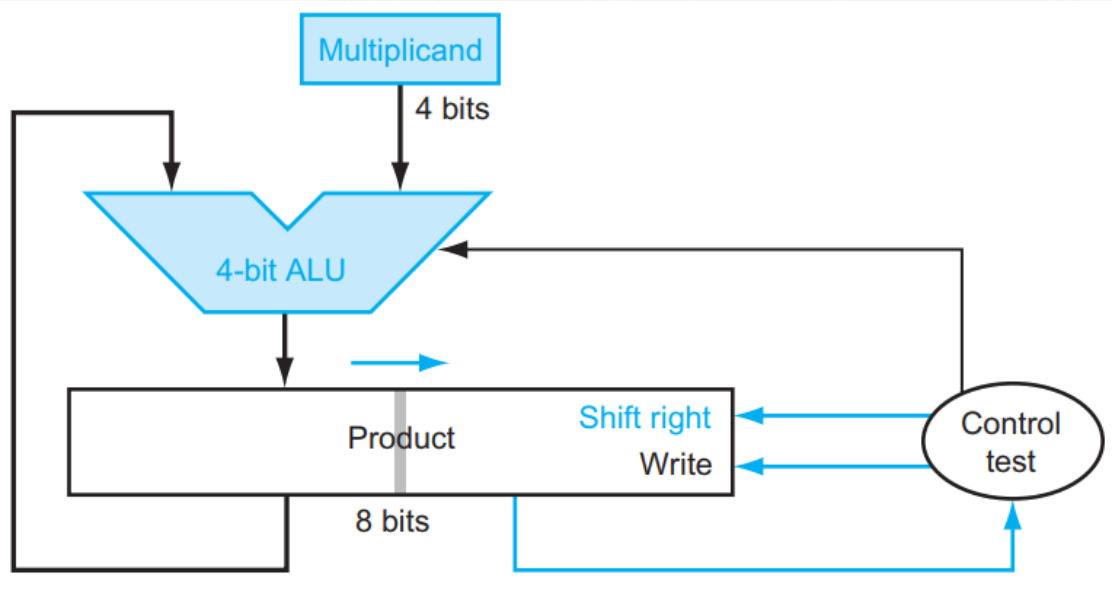


iter	Multiplicand	Product	Operation
ini	0010	0000 0111	
1	0010	0010 0111	**1**: Prod left half accumulate Shift right Prod
	0010	0001 0011	
2	0010	0011 0011	**1**: Prod left half accumulate Shift right Prod
	0010	0001 1001	
3	0010	0011 1001	**1**: Prod left half accumulate Shift right Prod
	0010	0001 1100	
4	0010	0000 1110	**0**: Shift right Prod
		res=00001110	done

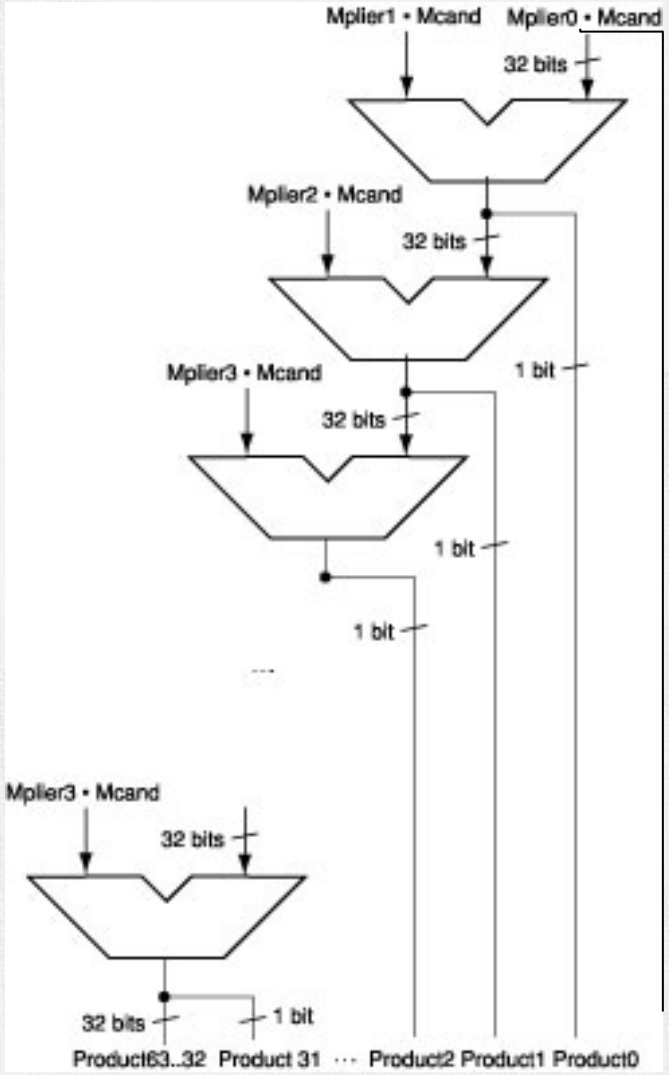
zhangj4@sustech.edu.cn

Optimized Multiplier

- Exercise: Multiply 8_{ten} (1000_{two}) by 11_{ten} (1011_{two}) :
- Values in Product register?
- How about Multiplicand register?



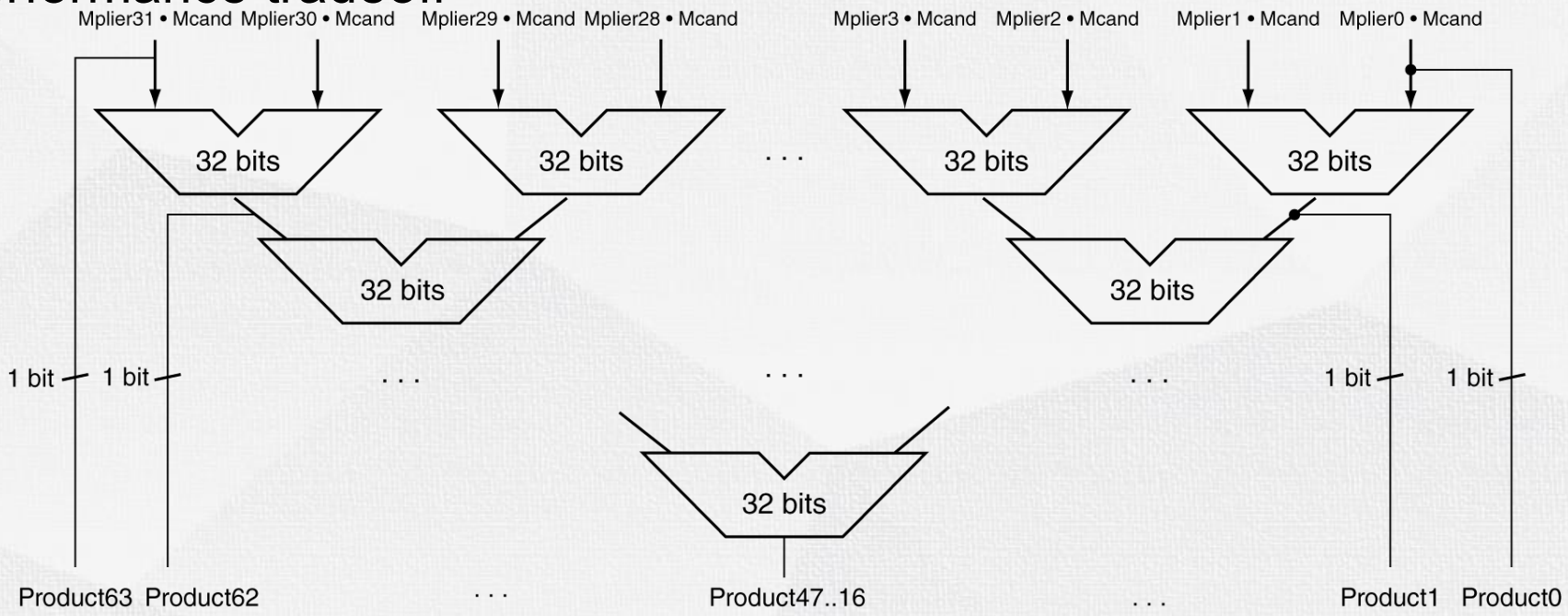
Faster Multiplier



- The previous algorithm requires a clock to ensure that the earlier addition has completed before shifting
- This algorithm can quickly set up most inputs – it then has to wait for the result of each add to propagate down – faster because no clock is involved
- High transistor cost

Faster Multiplier

- Uses multiple pipelined adders
  - Cost/performance tradeoff



- Can be pipelined
  - Several multiplication performed in parallel

Multiplication for Signed Integers

- Previously, we assume both the multiplicand and multiplier are positive integers. How about the multiplication for signed integers?
- Procedure: convert to positive number, calculate the product, then change the sign
- E.g. -7 ✕ -6
- |-7|=7: 0111_2, |-6|=6: 0110_2, 7 ✕ 6 = 0111_2 ✕ 0110_2 = 101010_2 = 42
- So, -7 ✕ -6 = 42
- Be noted that 4-bit multiplicand and multiplier can both represent an unsigned number in range [0, 15] and a signed number in range [-8, 7]
- It can also be calculated by thinking -7 as 11111001 and -6 as 11111010 and extend the sign of the product for signed numbers when we do shifting during the multiplication.

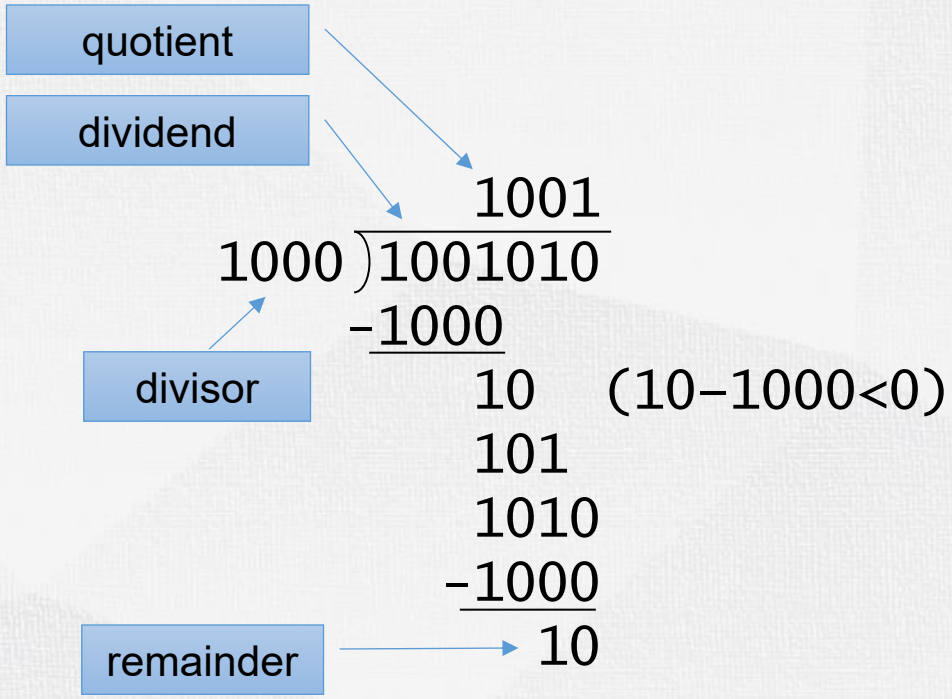
RISC-V Multiplication

- Four multiply instructions:
  - mul: multiply
    - Gives the lower 32 bits of the product
  - mulh: multiply high
    - Gives the upper 32 bits of the product, assuming the operands are signed
  - mulhu: multiply high unsigned
    - Gives the upper 32 bits of the product, assuming the operands are unsigned
  - mulhsu: multiply high signed/unsigned
    - Gives the upper 32 bits of the product, assuming one operand is signed and the other unsigned

Integer Division

zhangj4@sustech.edu.cn

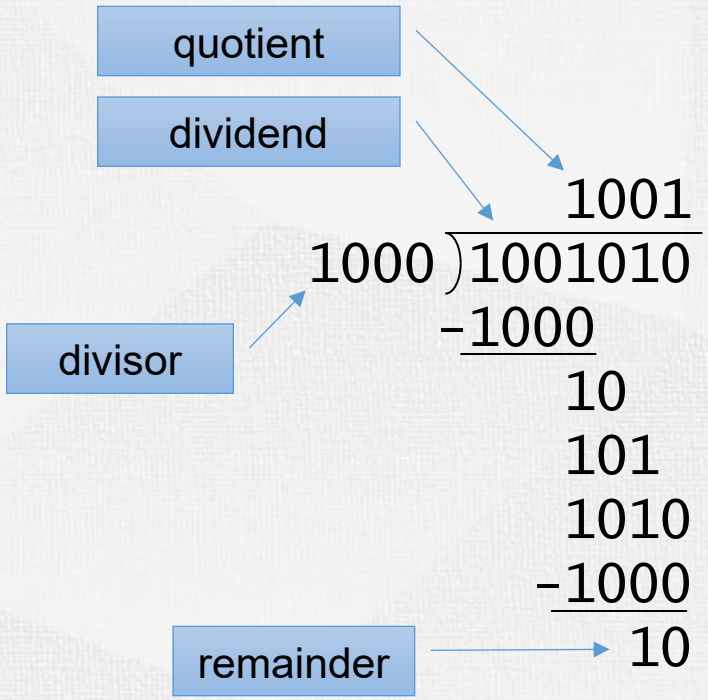
Division

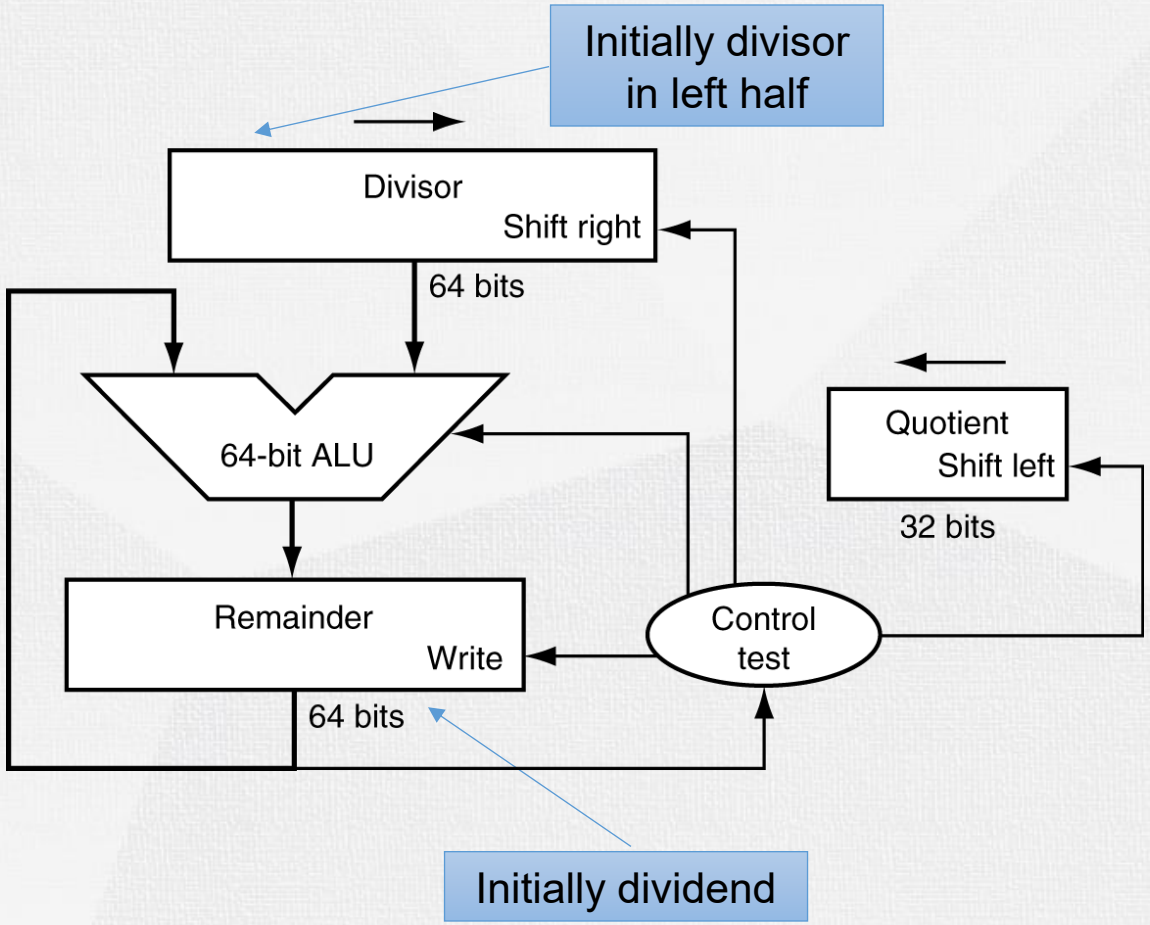


n-bit operands yield n-bit quotient and remainder

- Check for 0 divisor
- Long division approach
  - If divisor ≤ dividend bits
    - 1 bit in quotient, subtract
  - Otherwise
    - 0 bit in quotient, bring down next dividend bit
- Restoring division
  - Do the subtract, and if remainder goes < 0, add divisor back
- Signed division
  - Divide using absolute values
  - Adjust sign of quotient and remainder as required

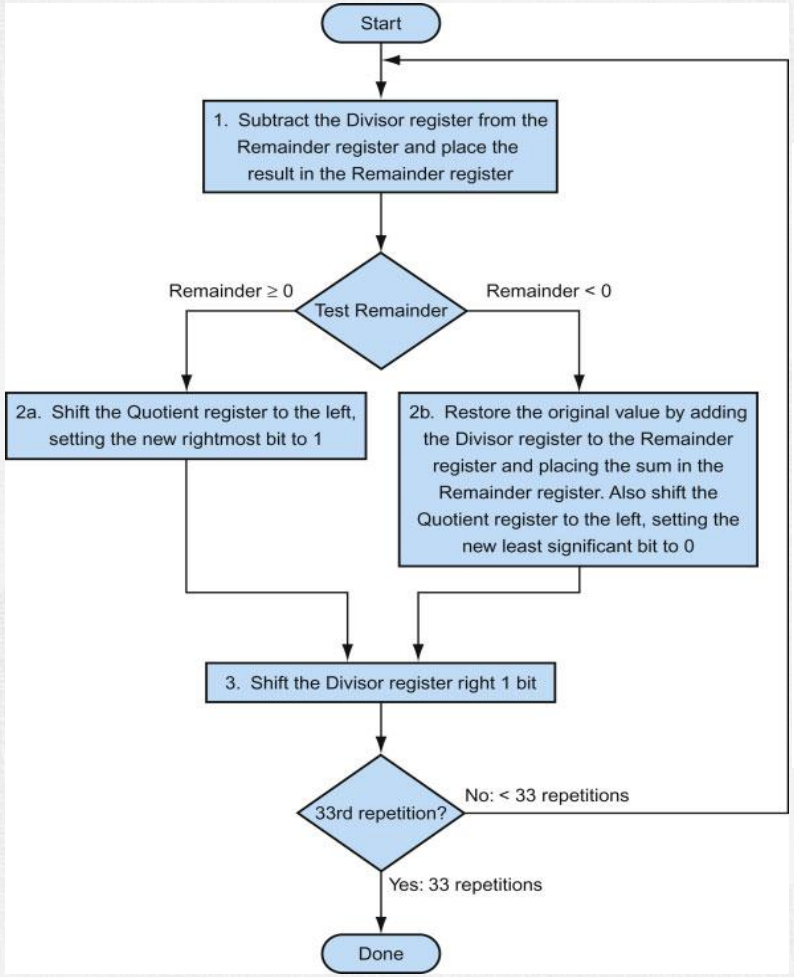
Division Hardware

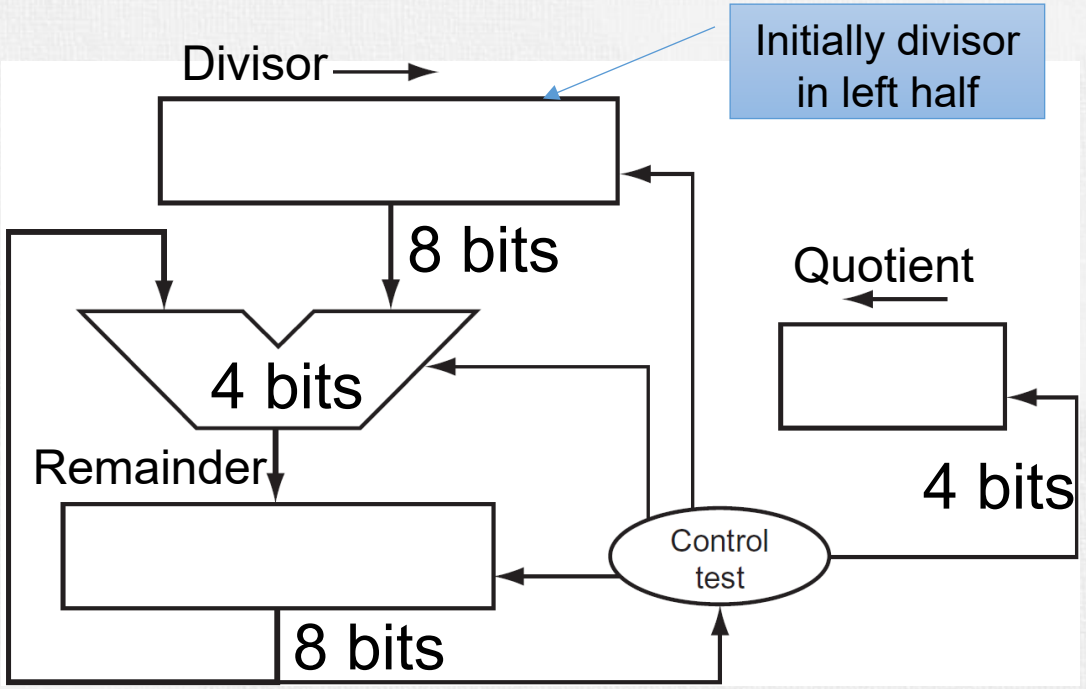




Division Example

- Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})





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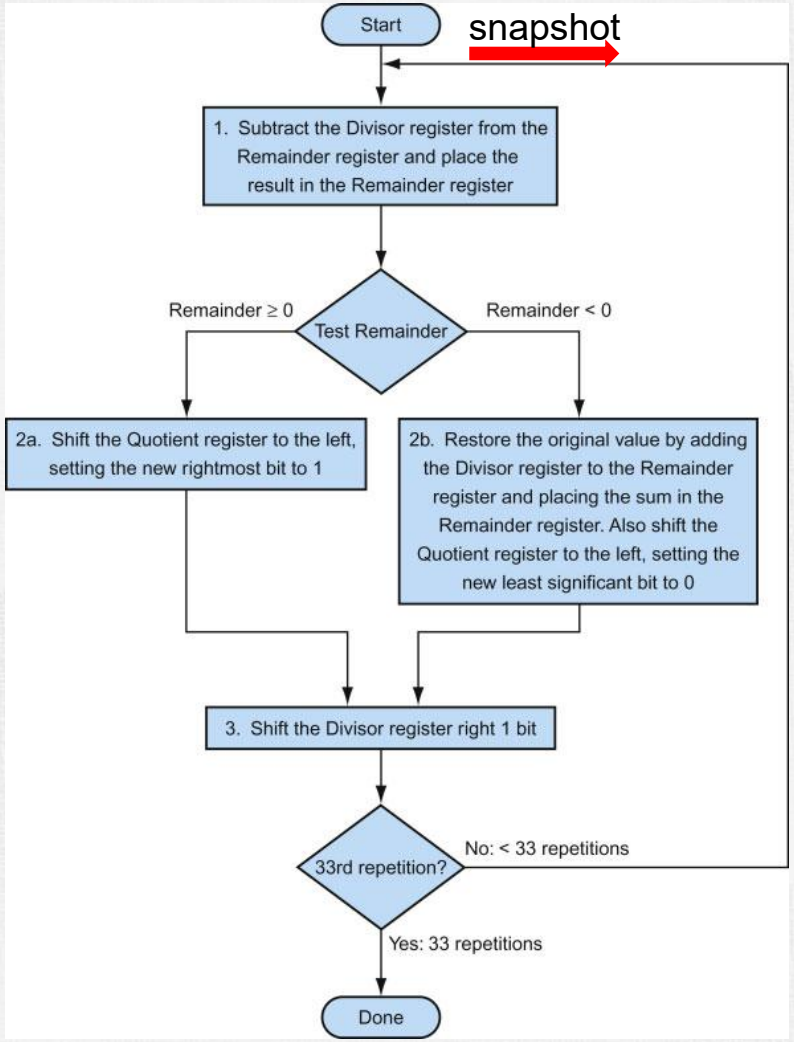

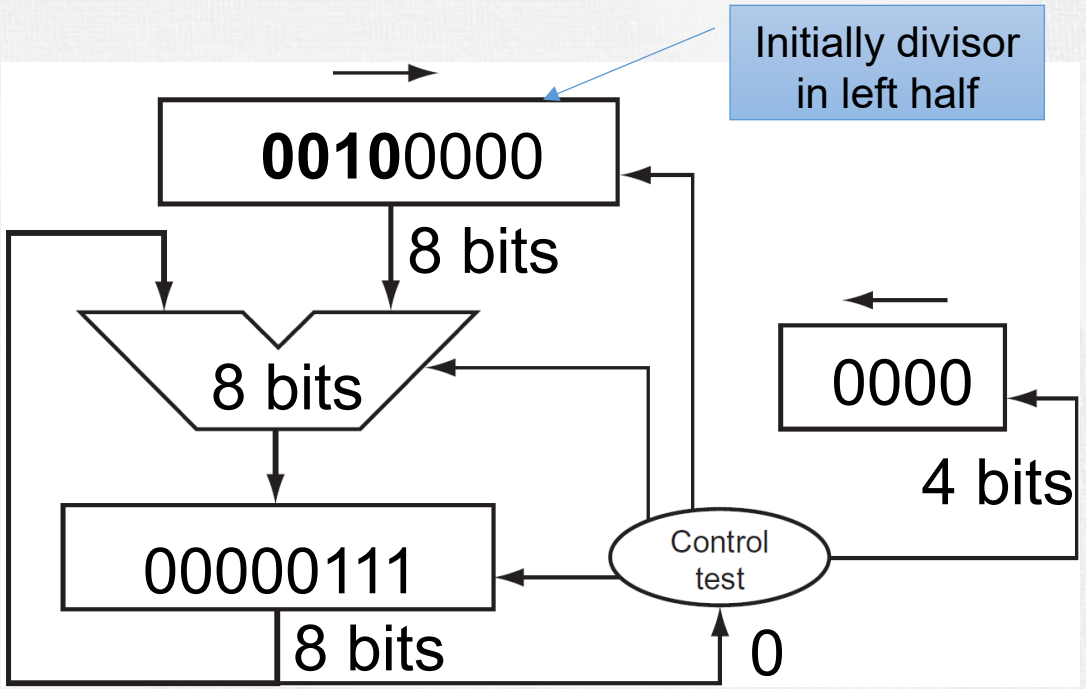

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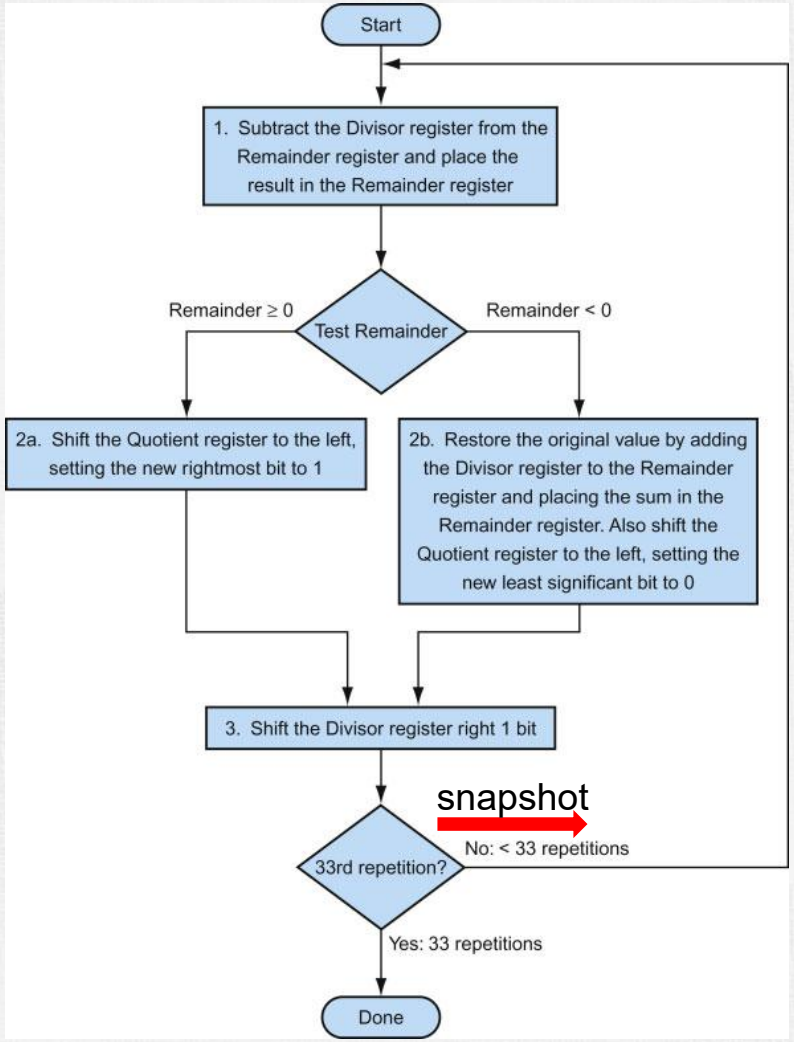
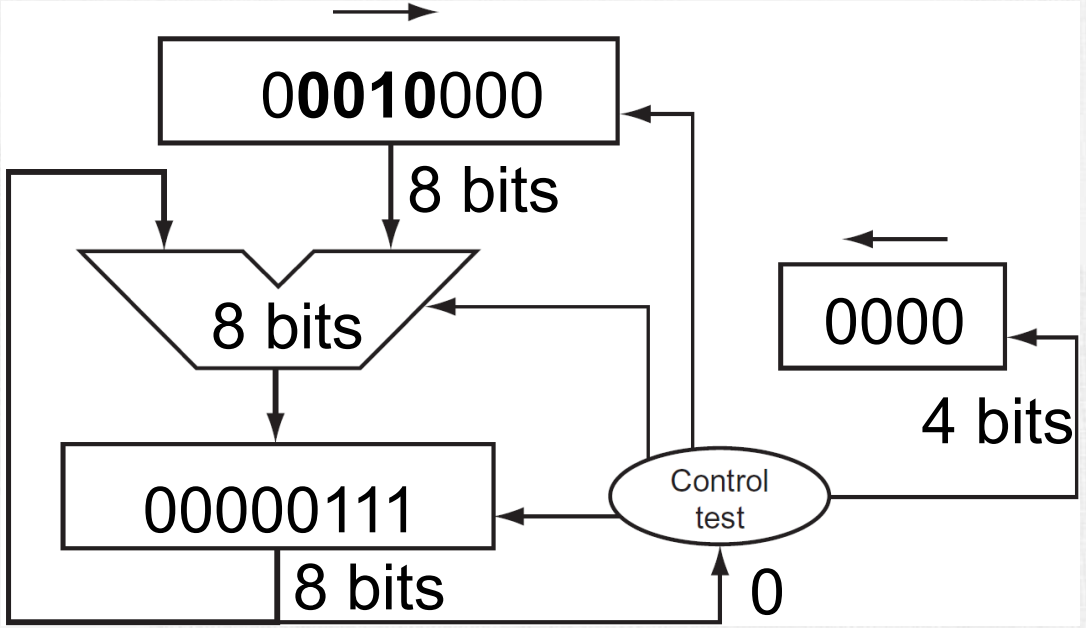
- Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})





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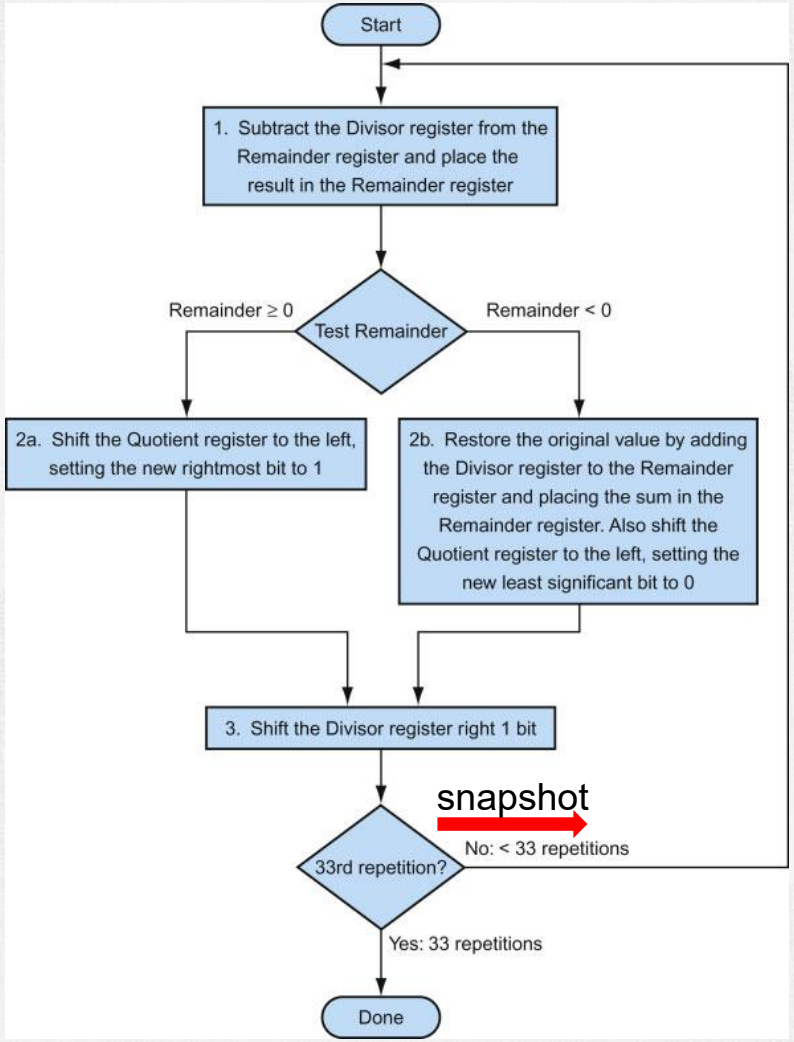
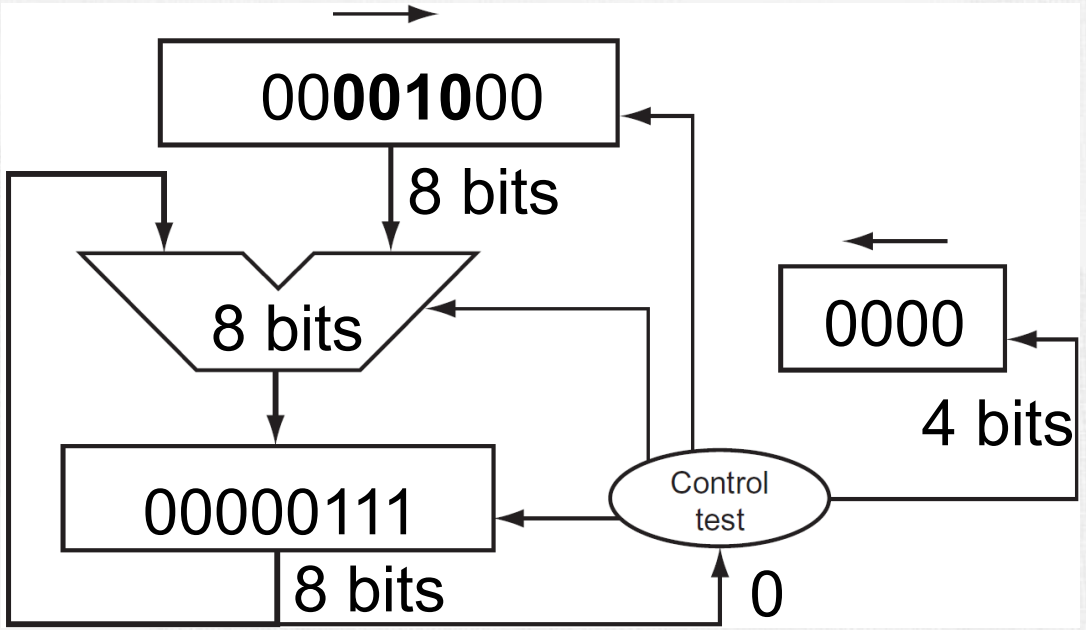
Division Example

- Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})





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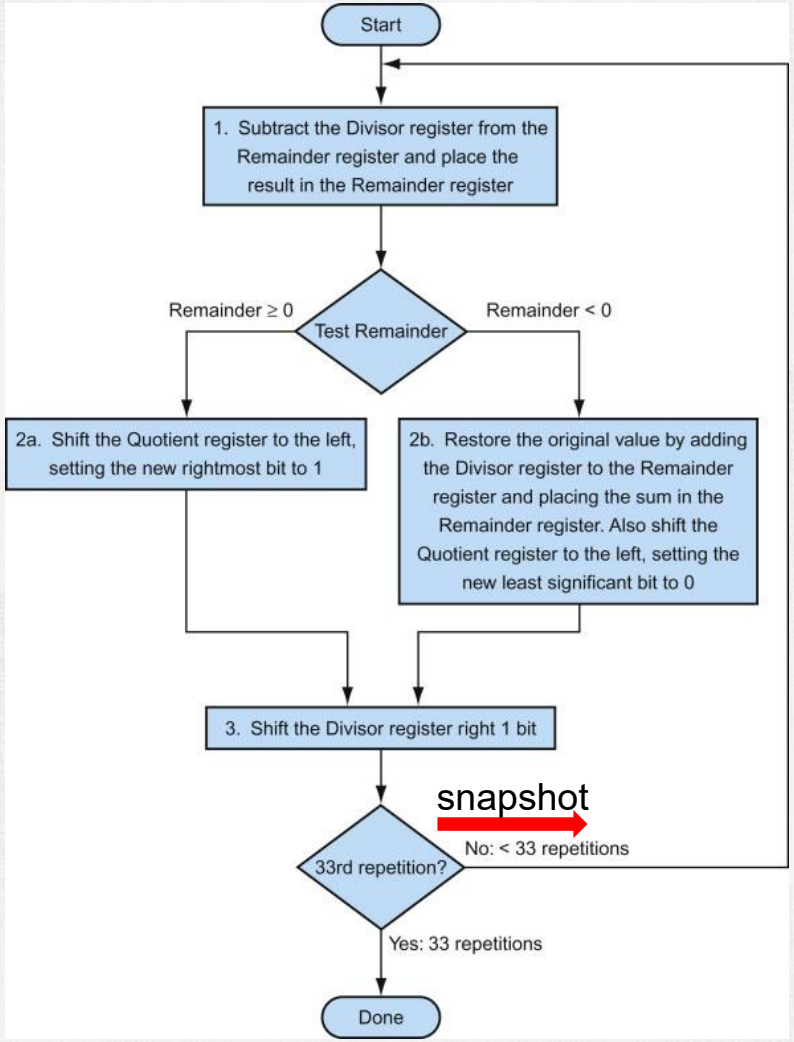
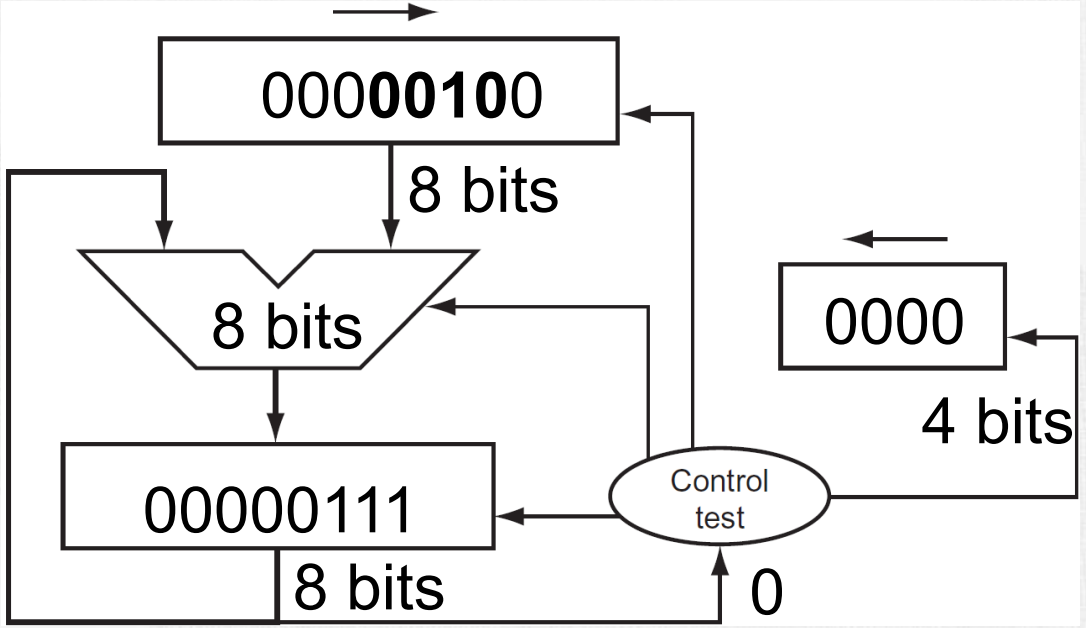
Division Example

- Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})





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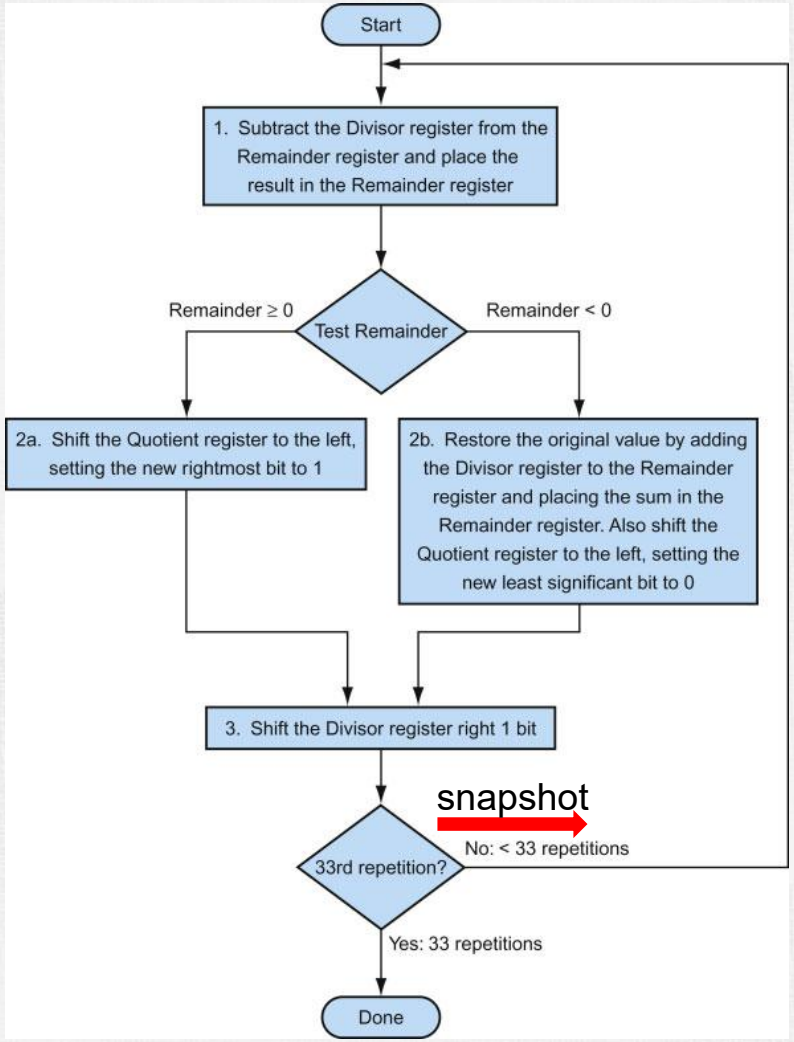
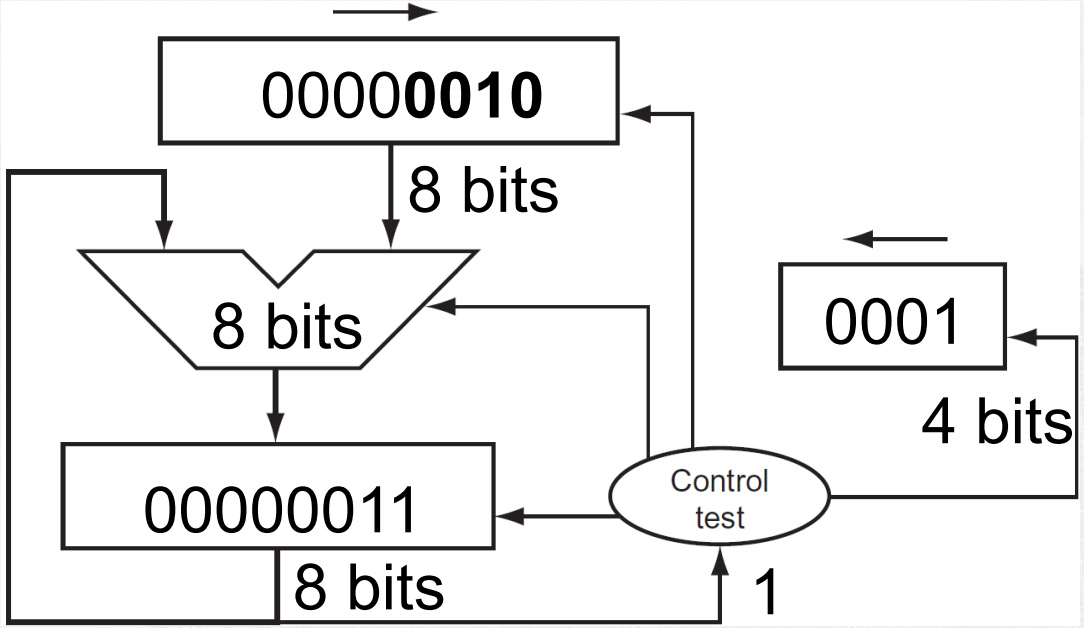
Division Example

- Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})





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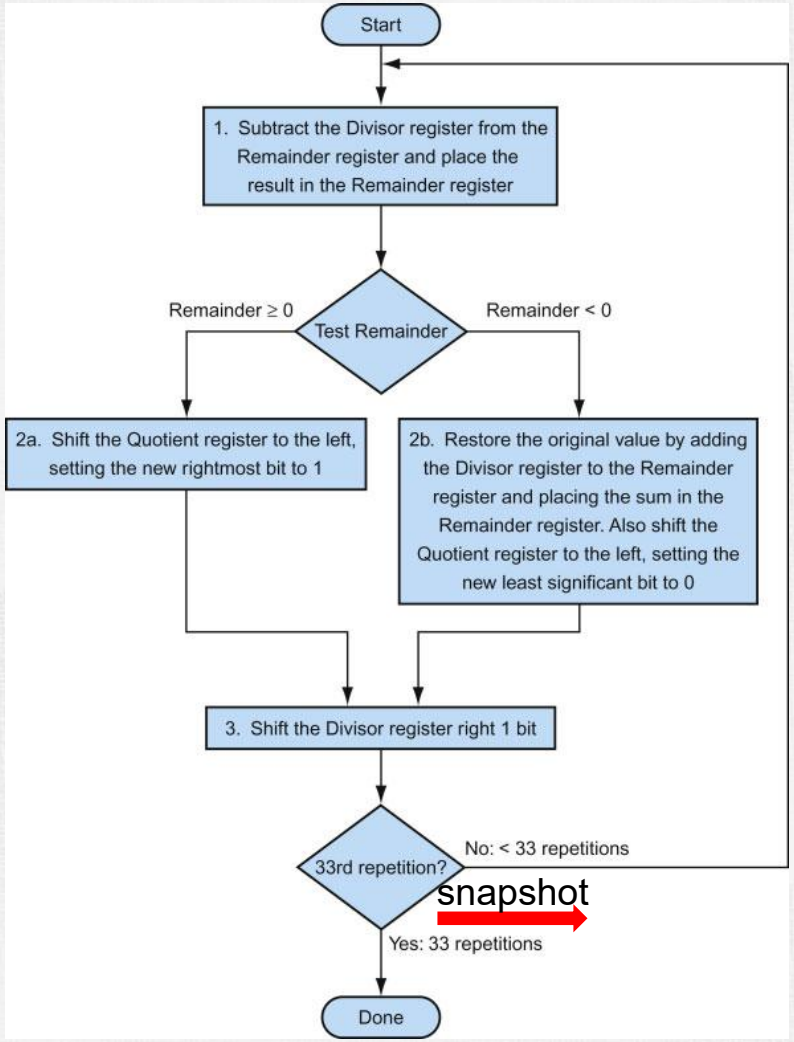
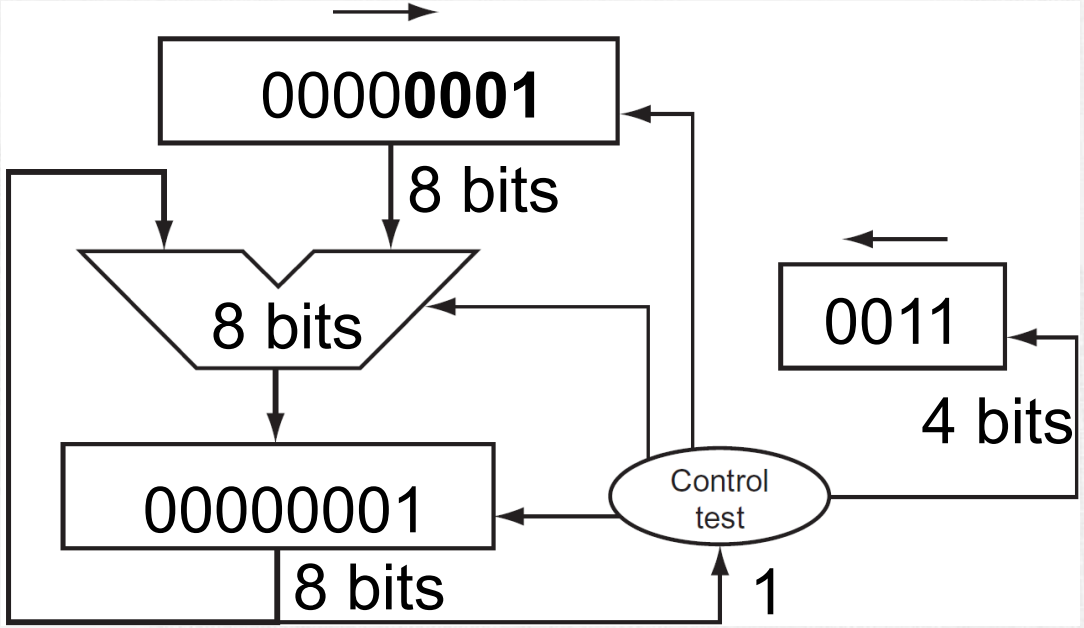
Division Example

- Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})





snapshot

Done!

zhangj4@sustech.edu.cn

Division Example

Divide 7_{ten} ($0000\ 0111_{two}$) by 2_{ten} (0010_{two})

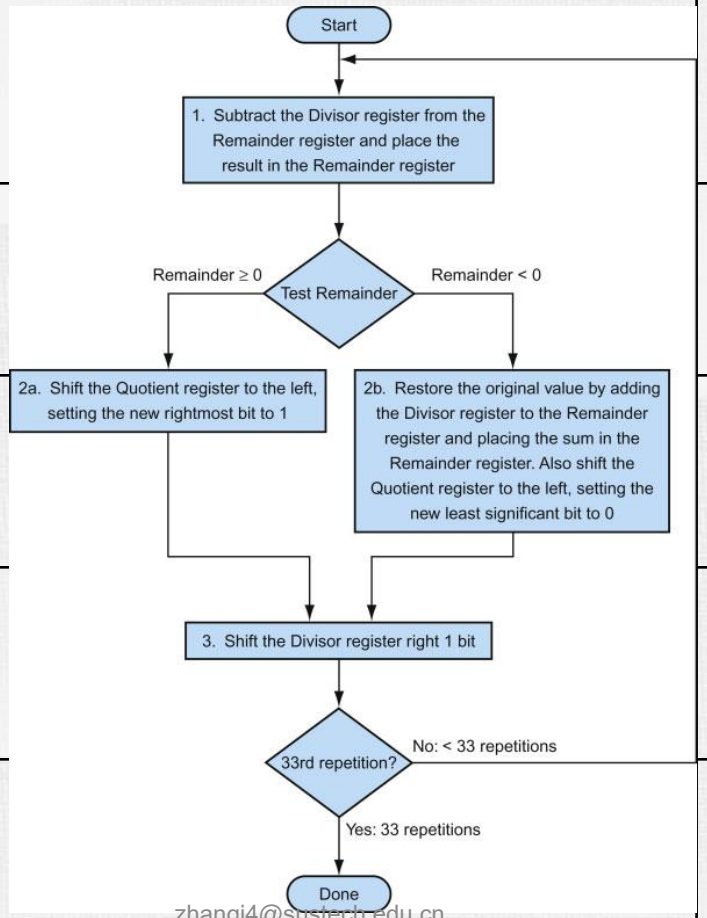
Iter	Step	Quot	Divisor	Remainder
0	Initial values	0000	0010 0000	0000 0111
1	Rem = Rem – Div Rem < 0 ➔ +Div, shift 0 into Q Shift Div right	0000 0000 0000	0010 0000 0010 0000 0001 0000	1110 0111 0000 0111 0000 0111
2	Same steps as 1	0000 0000 0000	0001 0000 0001 0000 0000 1000	1111 0111 0000 0111 0000 0111
3	Same steps as 1	0000 0000 0000	0000 1000 0000 1000 0000 0100	1111 1111 0000 0111 00000111
4	Rem = Rem – Div Rem >= 0 ➔ shift 1 into Q Shift Div right	0000 0001 0001	0000 0100 0000 0100 0000 0010	0000 0011 0000 0011 0000 0011
5	Same steps as 4	0001 0011 0011	0000 0010 0000 0010 0000 0001	0000 0001 0000 0001 0000 0001

8-bit dividend/4-bit divisor

Divide 74_{ten} (1001010_{two}) by 8_{ten} (1000_{two})

$$
\begin{array}{r}
1001 \\
1000\overline{)1001010} \\
-1000 \\
\hline
10 \\
101 \\
1010 \\
-1000 \\
\hline
10
\end{array}
$$

Iter	Step	Quot	Divisor	Remainder
0	Initial values	0000	10000000	01001010
1	1.1 1.2b 1.3	0000 0000 0000	10000000 10000000 01000000	11001010 01001010 01001010
2	2.1 2.2a 2.3	0000 0001 0001	01000000 01000000 00100000	00001010 00001010 00001010
3	3.1 3.2b 3.3	0001 0010 0010	00100000 00100000 00010000	11100101 00001010 00001010
4	4.1 4.2b 4.3	0010 0100 0100	00010000 00010000 00001000	11111010 00001010 00001010
5	5.1 5.2b 5.3	0100 1001 1001	00001000 00001000 00000100	00000010 00000010 00000010



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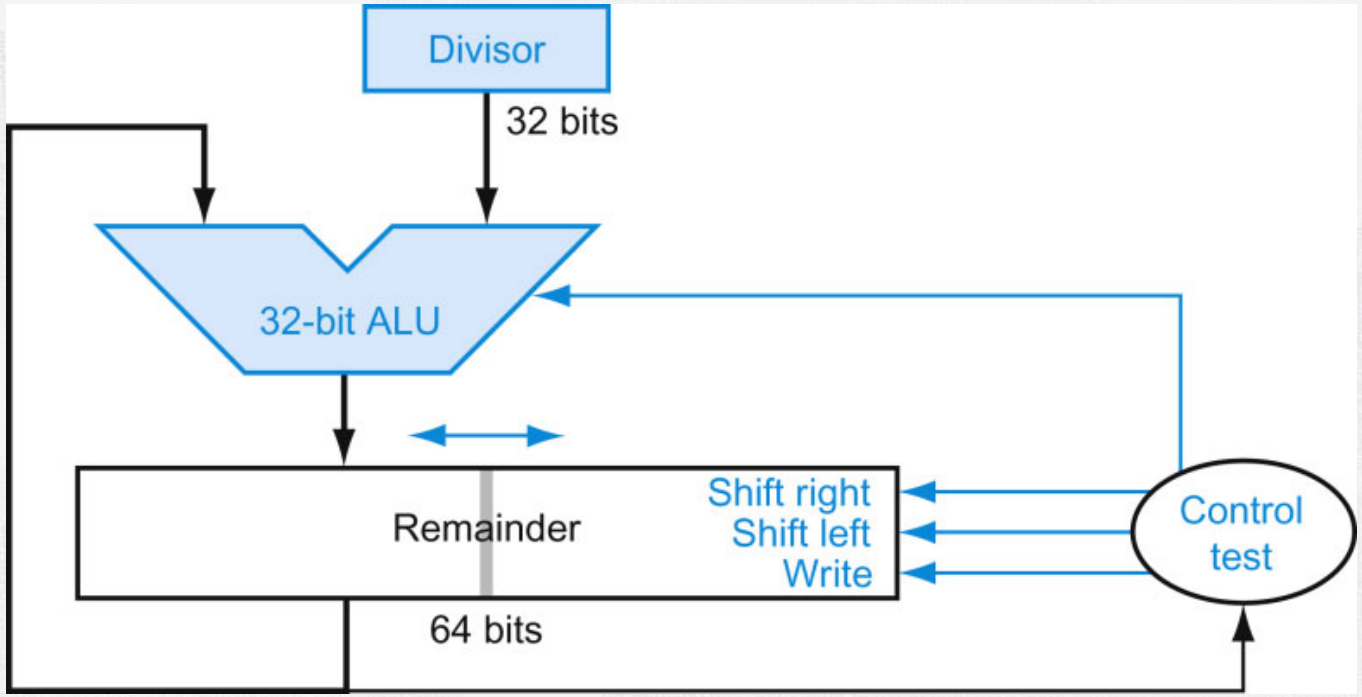
Optimized Divider

- One cycle per partial-remainder subtraction
- Looks a lot like a multiplier!
  - Same hardware can be used for both



Signed Division

- Convert to positive and adjust sign later
- Note that multiple solutions exist for the equation:
  Dividend = Quotient x Divisor + Remainder

+7 div +2 Quo = +3 Rem = +1
-7 div +2 Quo = -3 Rem = -1

Why not -7 div +2 Quo = -4 Rem = +1?
If so, -(x div y) != (-x) div y → programming challenge!

+7 div -2 Quo = -3 Rem = +1
-7 div -2 Quo = +3 Rem = -1

- Convention:
  - Dividend and remainder have the same sign
  - Quotient is negative if signs disagree
  - These rules fulfill the equation above

zhangj4@sustech.edu.cn

Faster Division

- Can’t use parallel hardware as in multiplier
  - Subtraction is conditional on sign of remainder
- Faster dividers (e.g. SRT devision) generate multiple quotient bits per step
  - Still require multiple steps

RISC-V Division

- Four instructions:
  - div, rem: signed divide, remainder
  - divu, remu: unsigned divide, remainder

- Overflow and division-by-zero don’t produce errors
  - Just return defined results
  - Faster for the common case of no error